Trace-characterising inequalities fail with the Michelson contrast of the functional and of the operators, and projections detect the failure

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Abstract

Many inequalities that hold for the trace characterise it: a positive functional that satisfies one of them for all operators is a trace. We study 25 such characterisations from the literature, among them Gardner's inequality $|\varphi(X)| \le \varphi(|X|)$, the triangle inequality for $\varphi(|\cdot|)$, monotonicity, Young's, Araki–Lieb–Thirring's, Peierls–Bogoliubov's and determinant inequalities, and inequalities for pairs of projections, and ask how strongly they fail for a non-tracial functional $\varphi = \mathrm{Tr}(W\,\cdot)$ on M_n. Our main observation is common to all of them: the violation is governed by Michelson contrasts. It vanishes when the density W or the operators entering the inequality have contrast zero, and it rises with both. For Gardner's inequality and its relatives we prove that the sharp constant depends on the two contrasts only through their product, $|\varphi(X)| \le (1-\Delta(W)^2\Delta(|X|)^2)^{-1/2}\varphi(|X|)$ in even dimension, with an explicit value in odd dimension. With the operators free, we determine the envelope as a function of $\delta = \Delta(W)$ exactly in sixteen cases and give lower bounds in four; the envelopes vanish at the trace to the orders δ, δ^2, δ^4 and δ^8, and the sharp constants are the Kantorovich constant, the condition number and the Specht ratio of W. On a direct sum $\bigoplus_c M_{k_c}$ only the contrast inside the blocks counts. A certified numerical search over all operators shows how the largest violation is attained: at projections for most characterisations, at two-level operators $\ell + (1-\ell)P$ for the others, so that in practice it suffices to test an inequality on projections and their affine images. Random sampling confirms both theses from the other side: random operators rarely violate the inequalities, and for monotonicity, subadditivity and the Powers–Størmer inequality essentially never, while samples of scaled projections violate them many times more often. The search also exhibits the exceptions: a threshold in the contrast of the operators below which no violation occurs (monotonicity, subadditivity, the spectral geometric mean), a peak at the contrast of the density (the determinant inequality), invariance under shifts (Peierls–Bogoliubov), and two inequalities that in dimension three are violated more strongly by operators that are not two-level. Finally, we determine the sharp bound of the unnormalised violation $|\mathrm{Tr}(XZ)| - \mathrm{Tr}(X|Z|)$ with both contrasts prescribed, which answers a question raised in our earlier work on the Michelson contrast of operators. Since the operators enter the inequalities through composites, we also describe how the contrast is transformed: under congruences, products and geometric means the rapidities $\operatorname{artanh}\Delta$ add, under Jordan products the angles $\arcsin\Delta$ add (and $AB + BA \ge 0$ for all relative positions if and only if $\Delta(A)^2 + \Delta(B)^2 \le 1$, Gustafson's criterion), and the commutator characterisation $\varphi(AB) = \varphi(BA)$ fails exactly by $2\Delta(W)\Delta(A)\Delta(B)$ in even dimension, with suitable normalisation.

Keywords: trace characterisation; positive functional; Michelson contrast; Kantorovich inequality; Specht ratio; two projections; two-level operators; von Neumann algebra.
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1 Introduction

Let φ be a positive linear functional on a C*-algebra. Gardner [18] proved that φ is a trace, $\varphi(XY) = \varphi(YX)$, if and only if

$$|\varphi(X)| \le \varphi(|X|) \qquad \text{for all } X. \tag{1}$$

This started a long series of results of the same form: an inequality that holds for every trace holds for all operators only if φ is a trace. On M_n or among the positive normal functionals on a von Neumann algebra, traces are characterised by the triangle inequality for $\varphi(|\cdot|)$ [18, 32], by $|\varphi(A^k)| \le \varphi(|A|^k)$ [29], by monotonicity $0 \le A \le B \Rightarrow \varphi(A^p) \le \varphi(B^p)$ [5, 30] and its weighted forms [23, 24], by Young's inequality [4, 16], by subadditivity of $\varphi \circ f$ for concave f [34], by the Peierls–Bogoliubov and Araki–Lieb–Thirring inequalities [9], by the generalised Powers–Størmer inequality [25], by inequalities for determinants and power means [12], by the spectral geometric mean [15], by inequalities for $|\varphi(A^2)|$ [14], by metrics on projections [11], by subadditivity on the projection lattice [29, Theorem 4], and by a family of inequalities in which only projections enter [6, 7, 8, 10]. Bikchentaev and Moslehian [13] survey these results; the canonical trace on M_n is characterised in [27, 12]. The dual problem, in which the functional is arbitrary and the operator is to be characterised as central, is treated in [28].

All these theorems answer a yes/no question. Here we ask a quantitative one: *how strongly does a trace-characterising inequality fail for a given non-tracial functional?* On M_n every positive functional has the form $\varphi_W = \mathrm{Tr}(W\,\cdot\,)$ with a density $W \ge 0$, and φ_W is a trace if and only if W is a scalar multiple of 1. The natural measure of the distance of W from the scalars that is invariant under scaling is the Michelson contrast

$$\Delta(W) = \frac{\lambda_{\max}(W) - \lambda_{\min}(W)}{\lambda_{\max}(W) + \lambda_{\min}(W)} \in [0, 1],$$

studied for operators in [1]. There, a Monte Carlo study [1, Sect. 5.1, Fig. 1] plotted the violation of Gardner's inequality against $\Delta(W)$ for real 2×2 and 3×3 matrices and conjectured that such violations are bounded by functions of the contrasts [1, Sect. 7]. We make this precise. For an inequality $L \le R$ (with L, R depending on φ and on operator variables) the *violation envelope* is

$$V_n(\delta) = \sup\Big\{ \frac{L - R}{N} : \; W \ge 0 \text{ in } M_n,\; \Delta(W) = \delta,\; \text{all operator variables free} \Big\},$$

where $N > 0$ is a normalisation that makes the quotient invariant under the scaling of W and of the variables (mostly $N = R$). By the characterisation theorems, $V_n(\delta) > 0$ for $\delta > 0$; by construction $V_n(0) = 0$.

Two theses. We study the 25 inequalities of Table 1 and find two properties that hold for all of them, with exceptions that we identify.

(I) The violation is governed by contrasts. Besides the contrast δ of the density we prescribe the contrast δ' of the operators that enter the inequality (of their moduli, for operators that are not positive). If either is zero there is no violation (Proposition 2.4); the violation needs a functional far from a trace *and* operators far from scalars. For Gardner's inequality the envelope is a function of the product alone:

$$V_n(\delta, \delta') = \frac{1}{\sqrt{1 - \delta^2 \delta'^2}} - 1 \qquad (n \text{ even}),$$

with an explicit, slightly smaller value for odd n (Theorem 4.1), and the same bound holds for the triangle inequality and for $\varphi(A + A^*) \le 2\varphi(|A^*|)$. For the extended Young inequality the operators $A = B$ with two-level modulus give $4\delta\delta'/(1 + \delta'^2 - 2\delta\delta')$, linear in each contrast (Proposition 5.3).

For the remaining items Appendix A shows $V(\delta,\delta')$ from a certified search: it rises with δ' for most of them, but some (monotonicity, subadditivity, the spectral geometric mean, the Powers–Størmer and the power-mean inequality) are violated only by operators of contrast close to 1, the determinant inequality is violated most strongly when the contrast of A equals that of W, and the Peierls–Bogoliubov inequality, being invariant under $A\mapsto A+c$, $B\mapsto B+c$, does not see the contrast of the operators at all.

(II) Projections detect the failure. A positive operator of contrast δ' with only two eigenvalues is, up to a factor, $\ell'+(1-\ell')P$ with a projection P. We call it *two-level*. Whenever the violation depends convexly on the spectrum of a variable, its maximum is attained at two-level operators (Lemma 2.5); this is the mechanism behind all closed forms below. Numerically (Table 3), in M_2 the largest violation over all operators is attained, or approached, by (scaled) projections for 22 of the 25 items (for 11 of them the variables are projections by definition) and by two-level operators for the other three; in M_3 the same holds except for the spectral geometric mean and the generalised Powers–Størmer inequality, which at large contrast are violated more strongly by operators that are not two-level. In practice, an inequality characterising the trace can be tested on projections and their affine images. Random sampling (Section 10.2) shows why this matters: random operators violate subadditivity, monotonicity and the Powers–Størmer inequality in fewer than one sample in a thousand, scaled projections in 0.7 to 21% of the samples, and for Gardner's inequality the fraction rises from 4% to 19%.

Results with the operators free. Write $a\ge b>0$ for the extreme eigenvalues of W and

$$h=\frac{a}{b}=\frac{1+\delta}{1-\delta},\qquad \kappa=\frac{a+b}{2\sqrt{ab}}=\frac{1}{\sqrt{1-\delta^2}},\qquad \psi=\arcsin\delta,$$

the condition number, the Kantorovich constant and the contrast angle of W.

Theorem A. *The envelopes of the inequalities of Table 1 are as follows.*

(a) $V_n=\kappa-1$ *for Gardner's inequality, the triangle inequalities for* $\varphi(|\cdot|)$ *on operators and on projections,* $\varphi(A+A^*)\le 2\varphi(|A^*|)$, $|\varphi(A^2)|\le\varphi(A^*A)$, $\varphi(|P-Q|)\le\varphi(P+Q)$ *and* $\varphi(\operatorname{Re}PQ)\le\varphi(|PQ|)$, *in every dimension* $n\ge 2$ *(Theorem 3.1, Theorem 6.1).*

(b) $V_n=\kappa^2-1$ *for* $\varphi(PQP)\le\varphi(Q)$; $V_n=\frac{1}{2}\tan^2(\psi/2)$ *for* $\varphi(\operatorname{Re}PQ)\ge 0$; $V_n=\delta/(1-\delta)$ *for* $\varphi(P\vee Q)\le\varphi(P)+\varphi(Q)$; $V_n=\delta$ *for* $\varphi(P+Q)\le\varphi(1+P\wedge Q)$ *(n even); and* V_n *is the maximum of an explicit function of one variable,* $\frac{(e-1)^2}{e(e+1)}\delta^2+O(\delta^3)$, *for* $\varphi(e^{QPQ})\le\varphi(e^P)$ *(n even) (Theorem 6.1).*

(c) $V_n=h-1$ *for the extension of Young's inequality to arbitrary operators;* $V_2=s/(1+s)$, $s=\sin(\psi/2)$, *for* $\varphi(PQ+QP)\le 2\varphi(QPQ)$; *for the inequalities that characterise the* canonical *trace,* $V_2=1-\sqrt{1-\delta^2}$ *for the AM–GM inequality* $(\det A)^{1/n}\le\frac{1}{n}\varphi(A)$, *increasing with* n *to* $1-1/S(h)$ *with the Specht ratio* S, *and* $V_n=\sqrt{(\ell+n-1)/(n\ell)}-1$, $\ell=1/h$, *for* $\varphi(A^2)^{1/2}\le\varphi(A)$ *(Section 5).*

(d) $V_n\ge\kappa-1$ *for the spectral geometric mean inequality,* $V_n\ge\kappa^2-1$ *for the Araki–Lieb–Thirring inequality,* $V_n\ge\delta^4/16+O(\delta^5)$ *for monotonicity and* $V_n\ge\delta^8/4096\,(1+O(\delta^2))$ *for its weighted form with* $r=1$; *and* $V_n\le h-1$ *for the first ten items (Section 7).*

Thus the envelopes vanish at the trace to different orders, δ, δ^2, δ^4 and δ^8 (Table 4): some characterisations are far more sensitive than others, and two inequalities that are equivalent for traces may fail at different rates. For instance, $\varphi(P\vee Q)+\varphi(P\wedge Q)=\varphi(P)+\varphi(Q)$ for every trace,

yet the lattice subadditivity fails as $\delta/(1-\delta)$ and $\varphi(|P-Q|) \le \varphi(P+Q)$, with the same right-hand side, only as $\kappa - 1 \sim \delta^2/2$.

The second result concerns finite-dimensional von Neumann algebras $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}$. A positive functional φ_W on $\mathcal{M}$ is a trace if and only if W is central, i.e. scalar on every block, however different the scalars are. The violation therefore cannot depend on the global contrast of W:

Theorem B. *Let $W = \bigoplus_c W_c > 0$ in $\mathcal{M}$. For every inequality of Table 1 except 11, 16 and 24, the largest normalised violation for φ_W on $\mathcal{M}$ is at most the largest one for φ_{W_c} on the blocks M_{k_c}, with equality for items 1–10, 12–14, 17, 18, 21–23 and 25. In particular a density that varies only between the blocks produces no violation at all (Theorem 8.1).*

The third result returns to the unnormalised quantity of [1], in which both operators are normalised to norm one and both contrasts are prescribed:

Theorem C. *Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, $N = \sum_c \lfloor k_c/2 \rfloor \ge 1$. For $X \ge 0$ and Z in $\mathcal{M}$ with $\|X\| = \|Z\| = 1$, $\Delta(X) = \delta < 1$ and $\Delta(|Z|) = \delta'$,*

$$\sup\bigl(|\operatorname{Tr}(XZ)| - \operatorname{Tr}(X|Z|)\bigr) = \frac{N\,\delta^2\delta'^2}{(1+\delta)(1+\delta')},$$

and the supremum is attained (Theorem 4.3).

All values in Theorems A and C and in Theorem 4.1 are confirmed by a certified numerical search over all variables, which also gives the envelopes of the five inequalities without a closed form (Section 10).

What is new. The characterisation theorems are due to the cited authors, and the constants κ, h and the Specht ratio, as well as the inequalities behind them (Kantorovich's inequality in the form of Greub and Rheinboldt [19, 17], Specht's reverse AM–GM inequality [31]), are classical. The upper bounds of Theorem 3.1 are the Kantorovich inequality; what we add is that the resulting constants are the exact envelopes. We found no previous study of the envelopes; the closest work is our earlier paper [1], which plots the unnormalised Gardner violation, defines no envelope and proves no bound. A short account of part of these results appears as a use case in the description of the software used for the computations [26].

Organisation. Section 2 fixes the notation, lists the inequalities, introduces the contrast of the operators and collects the tools, among them the vertex principle. Section 4 treats Gardner's inequality with both contrasts (Theorem 4.1 and Theorem C); Sections 3, 5, 6 and 7 prove Theorem A, and Section 8 Theorem B. Section 9 describes the contrast of sums, congruences, means, Jordan products and commutators. Section 10 reports the numerical results on both theses, and Section 11 concludes.

2 Setting and tools

2.1 Functionals, contrast and envelopes

Throughout, Tr is the usual trace on $M_n = M_n(\mathbb{C})$, $|X| = (X^*X)^{1/2}$, $\|X\|_1 = \operatorname{Tr}|X|$, $\operatorname{Re} X = \frac{1}{2}(X + X^*)$, and $\varphi = \varphi_W = \operatorname{Tr}(W\,\cdot\,)$ with $W \ge 0$; φ is a trace iff $W \in \mathbb{C}1$. We write $a \ge b$ for the extreme eigenvalues of W, e_1, e_2 for corresponding unit eigenvectors, and $\delta = \Delta(W)$; unless stated otherwise $W > 0$, i.e. $\delta < 1$. The quotient $(L-R)/N$ is invariant under $W \mapsto tW$, $t > 0$, and the envelope $V_n(\delta)$ is the supremum over all W with $\Delta(W) = \delta$; for a fixed density we write

$V(W)$. We call a value of an envelope *attained* if the supremum is a maximum and *approached* otherwise. Variables called P, Q, R are orthogonal projections, $A, B \geq 0$ unless said otherwise, and X, Y arbitrary; $P \wedge Q$ and $P \vee Q$ are the projections onto $\operatorname{ran} P \cap \operatorname{ran} Q$ and onto $\operatorname{ran} P + \operatorname{ran} Q$.

	inequality $L \leq R$	variables	N	source
1	$\lvert\varphi(X)\rvert \leq \varphi(\lvert X\rvert)$	X	R	[18]
2	$\varphi(\lvert X+Y\rvert) \leq \varphi(\lvert X\rvert) + \varphi(\lvert Y\rvert)$	X, Y	R	[18, 32]
3	$\varphi(A + A^*) \leq 2\varphi(\lvert A^*\rvert)$	A arbitrary	R	[10]
4	$\lvert\varphi(A^2)\rvert \leq \varphi(A^*A)$	A arbitrary	R	[29, Thm. 1, $k=2$], [14]
5	$\varphi(\lvert P-R\rvert) \leq \varphi(\lvert P-Q\rvert) + \varphi(\lvert Q-R\rvert)$	P, Q, R	R	[11]
6	$\varphi(A \natural B) \leq \sqrt{\varphi(A)\varphi(B)}$	$A, B > 0$	R	[15]
7	$\varphi((B^{1/2}AB^{1/2})^2) \leq \varphi(BA^2B)$	A, B	R	[9, Lemma 2, $r=2$, $p=1$]
8	$\varphi(\lvert AB^*\rvert) \leq \frac{1}{2}\varphi(\lvert A\rvert^2) + \frac{1}{2}\varphi(\lvert B\rvert^2)$	A, B arbitrary	R	extension of [4]
9	$\varphi(PQ + QP) \leq 2\varphi(QPQ)$	P, Q	$\varphi(P) + \varphi(Q)$	[10]
10	$0 \leq A \leq B \Rightarrow \varphi(A^2) \leq \varphi(B^2)$	$A \leq B$	R	[5, 30]
11	$(\det A)^{1/n} \leq \frac{1}{n}\varphi(A)$, $\varphi(1) = n$	A	$(\det A)^{1/n}$	[12, Thm. 3(iii)]
12	$\varphi((A+B)^{1/2}) \leq \varphi(A^{1/2}) + \varphi(B^{1/2})$	A, B	R	[34, Cor. 1]
13	$\varphi(A + B - \lvert A-B\rvert) \leq 2\varphi(A^{1/4}B^{1/2}A^{1/4})$	A, B	$\varphi(A+B)$	[25, Lemma 3.1]
14	$\lvert\varphi(PQ)\rvert \leq \varphi(P)^{1/3}\varphi(Q)^{2/3}$	P, Q	R	[8, Thm. 2(i)]
15	$\varphi(e^{P+Q}) \leq \varphi(e^{P/2}e^Q e^{P/2})$	P, Q	R	[8, Thm. 2(iv)]
16	$\varphi(e^A)\exp\frac{\varphi(e^{A/2}Be^{A/2})}{\varphi(e^A)} \leq \varphi(e^{A+B})$	A, B self-adj.	R	[9, Lemma 1]
17	$\varphi(PQP) \leq \varphi(Q)$	P, Q	R	[7, Thm. 3.4(iv)]
18	$\varphi(\lvert P-Q\rvert) \leq \varphi(P+Q)$	P, Q	R	[7, Thm. 3.4(v)]
19	$\varphi(P+Q) \leq \varphi(1 + P \wedge Q)$	P, Q	R	[7, Thm. 3.4(iii)]
20	$\varphi(e^{QPQ}) \leq \varphi(e^P)$	P, Q	R	[8, Thm. 2(v)]
21	$0 \leq \varphi(\operatorname{Re} PQ)$	P, Q	$\varphi(P) + \varphi(Q)$	[8, Thm. 2(vi)]
22	$\varphi(\operatorname{Re} PQ) \leq \varphi(\lvert PQ\rvert)$	P, Q	R	[8, Thm. 2(vii)]
23	$\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$	P, Q	R	[29, Thm. 4]
24	$\varphi(A^2)^{1/2} \leq \varphi(A)$, $\varphi(1) = n$	A	R	[12, Thm. 3(vii)]
25	$0 \leq A \leq B \Rightarrow \varphi(A^4) \leq \varphi(AB^2A)$	$A \leq B$	R	[24, Thms. 4, 5, $r=1$], [23]

Table 1: The inequalities studied, with the normalisation N of the violation $(L-R)/N$ and the source of the characterisation. Projections are P, Q, R; positive operators A, B unless stated otherwise. Remarks on the forms are in Section 2.2.

2.2 Remarks on the forms

Several sources state families of inequalities; we use one member. In item 4, $\lvert\varphi(A^k)\rvert \leq \varphi(\lvert A\rvert^k)$ for all A and one fixed $k \geq 1$ characterises the trace on M_n [29, Theorem 1]; $k = 1$ is Gardner's inequality. Item 7 is the Araki–Lieb–Thirring inequality $\varphi((B^{1/2}AB^{1/2})^{rp}) \leq \varphi((B^{r/2}A^rB^{r/2})^p)$ with $r = 2$, $p = 1$ [9, Lemma 2]. The published form of item 8 is $\varphi(\lvert AB\rvert) \leq \frac{1}{p}\varphi(A^p) + \frac{1}{q}\varphi(B^q)$ for $A, B \geq 0$ [4]; we use $p = q = 2$ and allow arbitrary A, B, in the form of Ando's matrix Young inequality [2], and Proposition 5.1 refers to this extension. Item 9 is the member $p = 1$ of the family $\varphi(PQ + QP) \leq 2\varphi((QPQ)^p)$, $p \in (0, 1]$ [10]. Items 10 and 25 are the cases $r = 0$ and $r = 1$ of $\varphi(A^{2r+2}) \leq \varphi(A^rB^2A^r)$ for $0 \leq A \leq B$, which characterises traces for every fixed $r \geq 0$ [24, Lemma 1, Theorems 4, 5]; on M_n the case $r = 0$ is in [5, 30] and $r > 0$ in [23, Theorem 6]. Items 11 and 24 characterise the canonical trace among the positive functionals with $\varphi(1) = n$ [12, Theorem 3]; item 24 is (vii) there with $p = 2$, $q = 1$ and needs φ faithful. Item 12 is the case $f(t) = t^{1/2}$ of Tikhonov's characterisation by $\varphi(f(A + B)) \leq \varphi(f(A)) + \varphi(f(B))$ for concave f with $f(0) = 0$ [34, Corollary 1]. Item 13 is the generalised Powers–Størmer inequality $\varphi(A + B) - \varphi(\lvert A - B\rvert) \leq 2\varphi(f(A)^{1/2}g(B)f(A)^{1/2})$,

$g(t) = t/f(t)$, with $f(t) = t^{1/2}$ [25, Lemma 3.1, Remark 4]; with $N = R$ its envelope would be infinite (Remark 7.3), so we normalise by $\varphi(A+B)$. Item 14 is Hölder's inequality for projections with the exponent $\alpha = 3$; for $\alpha = 2$ it holds for every positive φ [8, remark after Theorem 2]. Item 16 is the Peierls–Bogoliubov inequality, stated in [9] for positive H, K; both sides scale by e^c under $A \mapsto A+c$, $B \mapsto B+c$, so the envelope over self-adjoint arguments is the same. Items 17–19 are [7, Theorem 3.4], announced in [6], and items 14, 15, 20–22 are [8, Theorem 2].

2.3 Tools

Lemma 2.1 (Kantorovich inequality [19]). *If $b \le W \le a$, $a \ge b > 0$, then $\omega(W^2)\,\omega(1) \le \kappa^2\,\omega(W)^2$ for every positive functional ω, where $\kappa = (a+b)/(2\sqrt{ab})$.*

Proof. $(a-W)(W-b) \ge 0$ gives $\omega(W^2) \le (a+b)x - ab\,y$ with $x = \omega(W)$, $y = \omega(1)$, and $((a+b)x - aby)y \le \kappa^2 x^2$ because the difference is $(\kappa x - \sqrt{ab}\,y)^2$. □

Lemma 2.2 (Two projections; Halmos [22]). *For projections P, Q on a finite-dimensional Hilbert space H, H is the orthogonal sum of subspaces of dimension 1 or 2 invariant under P and Q. On each two-dimensional one there is an orthonormal basis f, f' with $P = ff^*$ and $Q = qq^*$, $q = \cos\theta\, f + e^{i\chi}\sin\theta\, f'$, $0 < \theta < \pi/2$.*

The next lemma gives a general upper bound; it shows that no envelope of items 1–8 and 10 can exceed the order δ.

Lemma 2.3. *For items 1–8 and 10, $V(W) \le h - 1$.*

Proof. In each item $L \le aL_0$, where L_0 is the expression for the trace or the trace norm (e.g. $|\varphi(X)| \le a\|X\|_1$, $\varphi(|X+Y|) \le a\,\mathrm{Tr}\,|X+Y|$, $|\varphi(A^2)| \le a\|A^2\|_1$), $L_0 \le R_{\mathrm{Tr}}$ by the trace inequality (for item 4 $\|A^2\|_1 \le \|A\|_2^2$; for item 7 the Araki–Lieb–Thirring inequality; for item 6, $A\natural B$ is unitarily similar to $(A^{1/2}BA^{1/2})^{1/2}$, so $\mathrm{Tr}(A\natural B) = \|A^{1/2}B^{1/2}\|_1 \le (\mathrm{Tr}\,A\,\mathrm{Tr}\,B)^{1/2}$), and $R \ge bR_{\mathrm{Tr}}$. Hence $L/R \le a/b = h$. For item 8, $\varphi(|AB^*|) \le a\|A\|_2\|B\|_2 \le \frac{a}{2}(\|A\|_2^2 + \|B\|_2^2)$. □

2.4 Contrasts of the operator variables

The envelope $V_n(\delta)$ lets the operator variables range over all operators. To see how the violation depends on the operators themselves, we also prescribe their contrast. For a positive variable A we use $\Delta(A)$, for an arbitrary or self-adjoint variable X the contrast $\Delta(|X|)$ of its modulus (for the Peierls–Bogoliubov inequality, item 16, the variables are positive, as in [9]), and we let every operator variable of an item have the same contrast $\delta' \in [0,1]$:

$$V_n(\delta,\delta') = \sup\Big\{\frac{L-R}{N} :\ \Delta(W) = \delta,\ \text{every operator variable of contrast } \delta'\Big\}.$$

Projections other than 0 and 1 have contrast 1, so for items 5, 9, 14, 15 and 17–23 only $\delta' = 1$ occurs, and $V_n(\delta,1) \le V_n(\delta)$ with equality whenever the supremum over all operators is approached by singular ones. A positive operator of contrast δ' whose spectrum has only two points is, up to a positive factor,

$$A = \ell' + (1-\ell')P, \qquad \ell' = \frac{1-\delta'}{1+\delta'}, \qquad P \text{ a projection},\ P \ne 0,1;$$

we call such operators *two-level*. They are the affine images of projections, and as $\delta' \to 1$ they become scaled projections.

Proposition 2.4. *If every operator variable is a positive multiple of the identity (for arbitrary and self-adjoint variables: if every modulus is), then $L \le R$ in every item of Table 1 and for every positive φ. Hence $V_n(\delta, 0) \le 0$ for all items; and $V_n(0, \delta') \le 0$, since for $\delta = 0$ the functional is a multiple of the trace (the canonical trace in items 11 and 24, where $\varphi(1) = n$).*

Proof. Item by item, with $|X| = x$, $|Y| = y$ scalar and $X = xU$, $Y = yV$ with unitaries (symmetries for self-adjoint X, Y), and $A = a$, $B = b$ scalars: (1) $|\varphi(U)| \le \varphi(1)$; (2) $\varphi(|xU + yV|) \le \|xU + yV\|\varphi(1) \le (x + y)\varphi(1)$; (3) and (4) as (1); (6) and (7) are equalities; (8) $|AB^*| = xy$ and $xy \le \frac{1}{2}(x^2 + y^2)$; (10) and (25) follow from $a \le b$; (11) is an equality for $\varphi(1) = n$; (12) $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$; (13) $a + b - |a - b| = 2\min(a, b) \le 2\sqrt{ab}$; (16) is an equality; (24) $\varphi(a^2)^{1/2} = a\sqrt{n} \le an$. □

So a violation needs both contrasts: the functional must be far from a trace and the operators must be far from scalars. The next lemma is the mechanism behind the two-level operators: whenever the violation depends convexly on the spectrum of a variable, the extremal spectra have only two points.

Lemma 2.5 (Vertex principle)**.** *Fix all variables except the eigenvalues $\lambda \in [\ell', 1]^n$ of one positive variable (its eigenbasis fixed), and let $F(\lambda)$, $N(\lambda) > 0$ be the resulting values of $L - R$ and N. If $\lambda \mapsto F(\lambda) - cN(\lambda)$ is convex for every $c \in \mathbb{R}$, then the supremum of F/N over the spectra with $\max_i \lambda_i = 1$ and $\min_i \lambda_i = \ell'$ equals its supremum over the two-level spectra $\lambda \in \{\ell', 1\}^n$ (both values occurring).*

Proof. The admissible set is the union of the faces $\{\lambda_i = 1,\ \lambda_j = \ell'\}$ of the cube, and the vertices of these faces are the two-level spectra. If $F/N > c$ at an admissible point, then $F - cN > 0$ there, and by convexity $F - cN > 0$ at a vertex of the same face, where therefore $F/N > c$. □

The hypothesis holds for Gardner's inequality ($F - cN = |\varphi(UT)| - (1 + c)\varphi(T)$ for $X = UT$: the modulus of a linear function minus a linear one), for item 3, for item 24 ($\varphi(A^2)^{1/2} = \|W^{1/2}A\|_2$ is a norm of a linear function of λ) and for the unnormalised Gardner violation of Section 4; for these items two-level operators suffice. For the other items we test numerically whether they do (Section 10).

3 The Kantorovich class

Theorem 3.1. *Let $W > 0$ in M_n, $n \ge 2$, and $\varphi = \varphi_W$. For all X, Y, A and all projections P, Q, R,*

$$|\varphi(X)| \le \kappa\,\varphi(|X|), \qquad \varphi(A + A^*) \le 2\kappa\,\varphi(|A^*|), \qquad |\varphi(A^2)| \le \kappa\,\varphi(A^*A),$$
$$\varphi(|X + Y|) \le \kappa\big(\varphi(|X|) + \varphi(|Y|)\big), \qquad \varphi(|P - R|) \le \kappa\big(\varphi(|P - Q|) + \varphi(|Q - R|)\big).$$

All five constants are attained. Hence $V_n = \kappa - 1$ for items 1–5 in every dimension $n \ge 2$.

Proof. Let $X = U|X|$ be a polar decomposition and $T = |X|$. Then $\varphi(X) = \mathrm{Tr}(WUT) = \mathrm{Tr}\big((WT^{1/2})^* UT^{1/2}\big)$, so for every contraction U

$$|\varphi(X)|^2 \le \mathrm{Tr}(W^2T)\ \mathrm{Tr}\,T \le \kappa^2(\mathrm{Tr}\,WT)^2$$

by the Cauchy–Schwarz inequality and Lemma 2.1 for $\omega = \mathrm{Tr}(T\,\cdot\,)$. The second inequality follows in the same way from $\varphi(A + A^*) = 2\mathrm{Re}\,\varphi(|A^*|V)$, where $A = |A^*|V$ is a polar decomposition. For the fourth, write $X + Y = S|X + Y|$; then $\varphi(|X + Y|) = \varphi(S^*X) + \varphi(S^*Y)$, and each term is bounded as above with the contraction S^*U_X or S^*U_Y in place of U. The fifth is the fourth with $X = P - Q$,

$Y = Q - R$. For the third, put $X = AW^{1/2}$ in an eigenbasis of W with eigenvalues w_i. Then $\varphi(A^*A) = \sum_{ij} |X_{ij}|^2$ and

$$\varphi(A^2) = \sum_i X_{ii}^2 + \sum_{i<j} X_{ij}X_{ji}\frac{w_i + w_j}{\sqrt{w_i w_j}}, \qquad \frac{w_i + w_j}{\sqrt{w_i w_j}} \le 2\kappa,$$

so $|\varphi(A^2)| \le \sum_i |X_{ii}|^2 + \kappa \sum_{i<j}(|X_{ij}|^2 + |X_{ji}|^2) \le \kappa\varphi(A^*A)$.

Equality, with e_1, e_2 eigenvectors of W for a, b: in the first two inequalities for $T = vv^*$ with $|\langle e_1, v\rangle|^2 = \frac{b}{a+b}$, $|\langle e_2, v\rangle|^2 = \frac{a}{a+b}$ and $Uv = Wv/\|Wv\|$ (equality in Cauchy–Schwarz and in Lemma 2.1); in the third for $A = \sqrt{a}\, e_1 e_2^* + \sqrt{b}\, e_2 e_1^*$; in the fourth for $X = uu^*$, $Y = -u'u'^*$ with $u, u' = \sin t\, e_1 \pm \cos t\, e_2$ and $\tan t = \sqrt{b/a}$; in the fifth for $P = uu^*$, $Q = 0$, $R = u'u'^*$. □

Remark 3.2. For a fixed density, $\sup_X |\varphi(X)|/\varphi(|X|) = \kappa(W)$ exactly. In particular the normalised violation of Gardner's inequality depends on W only through its extreme eigenvalues, not on the dimension or on the rest of the spectrum.

4 Gardner's inequality with two contrasts

For Gardner's inequality both contrasts can be prescribed and the envelope found exactly. It depends on the two contrasts only through their product.

Theorem 4.1. *Let $W > 0$ and X be in M_n, $n \ge 2$, with $\Delta(W) = \delta$ and $\Delta(|X|) = \delta'$, $\delta, \delta' \in [0, 1]$, $\delta\delta' < 1$. Then*

$$|\varphi_W(X)| \le \frac{1}{\sqrt{1 - \delta^2\delta'^2}}\, \varphi_W(|X|),$$

and for even n the constant is attained: $V_n(\delta, \delta') = \kappa(\delta\delta') - 1 = (1 - \delta^2\delta'^2)^{-1/2} - 1$ for Gardner's inequality. For odd $n = 2k + 1$,

$$V_n(\delta, \delta') = \max_{0 \le \gamma \le 1} \frac{k\sqrt{S^2 + \gamma pq} + \ell\ell'}{k(S + \gamma p) + \ell\ell'} - 1, \qquad S = \ell + \ell',\ q = (1 + \ell)(1 + \ell'),\ p = (1 - \ell)(1 - \ell'),$$

with $\ell = \frac{1-\delta}{1+\delta}$, $\ell' = \frac{1-\delta'}{1+\delta'}$; it is smaller than $\kappa(\delta\delta') - 1$ for $\delta' < 1$ and tends to it as $n \to \infty$. For $\delta' = 1$ both give Theorem 3.1. The upper bound $\kappa(\delta\delta')$ holds also for items 2 and 3, with the contrasts of $|X|, |Y|$ and of $|A^*|$.

Proof. Write $X = UT$ with $T = |X|$. The supremum of $|\operatorname{Tr}(WUT)| = |\operatorname{Tr}(U\,TW)|$ over unitaries U is $\|TW\|_1$, so we must bound $\|TW\|_1 / \operatorname{Tr}(TW)$. For fixed eigenbases, $\|TW\|_1 - c\operatorname{Tr}(TW)$ is a convex function of the eigenvalues of T (a norm of a linear function minus a linear one), and likewise of the eigenvalues of W. By Lemma 2.5, applied to T and then to W, we may assume $W = a(\ell + (1 - \ell)P)$ and $T = t(\ell' + (1 - \ell')Q)$ with projections P, Q and $\ell = \frac{1-\delta}{1+\delta}$, $\ell' = \frac{1-\delta'}{1+\delta'}$. By Lemma 2.2, $\mathbb{C}^n$ splits into subspaces of dimension 1 or 2 invariant under P and Q, hence under W and T, and $\|TW\|_1$ and $\operatorname{Tr}(TW)$ are sums over them. On a one-dimensional summand the two contributions are equal. On a two-dimensional one, W and T have the eigenvalues x_1, x_2 and y_1, y_2, namely $a, a\ell$ and $t, t\ell'$; with $S = x_1y_2 + x_2y_1$, $q = (x_1 + x_2)(y_1 + y_2)$, $p = (x_1 - x_2)(y_1 - y_2)$ and γ the squared overlap of the leading eigenvectors, the proof of Proposition 4.2 gives $\|TW\|_1^2 = S^2 + \gamma pq$ and $\operatorname{Tr}(TW) = S + \gamma p$. The function $\gamma \mapsto (S^2 + \gamma pq)/(S + \gamma p)^2$ is maximal at $\gamma = S/q \in [0, 1]$, with the value

$$\frac{q^2}{4S(S + p)} = \frac{(x_1 + x_2)^2(y_1 + y_2)^2}{4(x_1y_2 + x_2y_1)(x_1y_1 + x_2y_2)} = \frac{1}{1 - \delta^2\delta'^2},$$

the last step by inserting $x_2/x_1 = \ell$, $y_2/y_1 = \ell'$. A quotient of sums is at most the largest quotient of the summands. Equality for even n: $n/2$ copies of the 2×2 configuration $W = \mathrm{diag}(1,\ell)$, T with the eigenvalues $1, \ell'$ at $\gamma = S/q$, and U the adjoint polar factor of TW. For odd n at least one summand is one-dimensional and contributes the same amount $m \geq \ell\ell'$ (normalising the largest eigenvalues of W and T to 1) to $\|TW\|_1$ and to $\mathrm{Tr}(TW)$; the quotient $(\sum_j L_j + m)/(\sum_j N_j + m)$ over at most k two-dimensional summands with $L_j \geq N_j$ is largest for k summands, $m = \ell\ell'$, and, by the argument of Lemma 2.5 applied to $L - cN$, the same γ in every summand, which gives the stated maximum; it is attained by $W = \mathrm{diag}(1,\ell)^{\oplus k} \oplus \ell$ and T correspondingly. Items 2 and 3 reduce to the first inequality as in the proof of Theorem 3.1. □

Thus the normalised violation of Gardner's inequality is $\frac{1}{2}\delta^2\delta'^2 + O(\delta^4\delta'^4)$: it vanishes to second order in each contrast and is symmetric in the density and the operator. The unnormalised violation studied in [1] behaves in the same way (Theorem 4.3 below).

In [1, Sect. 5] the violation of Gardner's inequality was studied without normalisation: for $X \geq 0$ with $\|X\| = 1$ (the density) and Z with $\|Z\| = 1$, the quantity

$$z(X,Z) = |\,\mathrm{Tr}(XZ)| - \mathrm{Tr}(X|Z|)$$

was sampled against $\Delta(X)$ and $\Delta(|Z|)$. Writing $Z = U|Z|$ with a unitary (orthogonal, in the real case) U, $z = |\,\mathrm{Tr}(XUY)| - \mathrm{Tr}(XY)$ with $Y = |Z| \geq 0$, and $\sup_U z = \|YX\|_1 - \mathrm{Tr}(XY)$ by the duality of the trace norm, the supremum being attained at the polar unitary of YX, which is real if X, Y are. We first solve the 2×2 case.

Proposition 4.2. *Let $X, Y \geq 0$ in $M_2(\mathbb{C})$ have eigenvalues $x_1 \geq x_2 \geq 0$ and $y_1 \geq y_2 \geq 0$, $x_1, y_1 > 0$. For every unitary U,*

$$|\,\mathrm{Tr}(XUY)| - \mathrm{Tr}(XY) \;\leq\; \frac{(x_1 - x_2)^2 (y_1 - y_2)^2}{4(x_1 + x_2)(y_1 + y_2)}.$$

Let $S = x_1 y_2 + x_2 y_1$, $q = (x_1 + x_2)(y_1 + y_2)$, and $\gamma = |\langle e_1, f_1\rangle|^2$ for unit eigenvectors e_1 of X for x_1 and f_1 of Y for y_1. If $x_1 > x_2$ and $y_1 > y_2$, the bound is attained iff $|\,\mathrm{Tr}(XUY)| = \|YX\|_1$ and $\gamma = \frac{1}{4} + \frac{S}{2q}$; it is attained by real X, Y, U.

Proof. Let $p = (x_1 - x_2)(y_1 - y_2)$, so that $q - 2S = p$. First, $|\,\mathrm{Tr}(XUY)| = |\,\mathrm{Tr}(YXU)| \leq \|YX\|_1$. Expanding in the eigenbases, $\mathrm{Tr}(XY) = \sum_{ij} x_i y_j |\langle e_i, f_j\rangle|^2 = S + \gamma p$, since the matrix $(|\langle e_i, f_j\rangle|^2)$ is doubly stochastic. The singular values of YX satisfy $s_1^2 + s_2^2 = \mathrm{Tr}(X^2Y^2)$ and $s_1 s_2 = |\det YX| = x_1 x_2 y_1 y_2$, so $\|YX\|_1^2 = \mathrm{Tr}(X^2Y^2) + 2x_1x_2y_1y_2 = S^2 + \gamma p q$. With $\sqrt{r} \leq r/q + q/4$ (equality iff $\sqrt{r} = q/2$),

$$\|YX\|_1 - \mathrm{Tr}(XY) \leq \frac{S^2 + \gamma p q}{q} + \frac{q}{4} - S - \gamma p = \frac{(q/2 - S)^2}{q} = \frac{p^2}{4q},$$

with equality iff $\gamma p q = q^2/4 - S^2 = \frac{p}{2}(\frac{q}{2} + S)$. Every $\gamma \in [0,1]$ is realised by a real rotation, and the polar unitary of a real YX is real. □

Theorem 4.3. *Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and $N = \sum_c \lfloor k_c/2 \rfloor \geq 1$. For $X \geq 0$ and $Y \geq 0$ in $\mathcal{M}$ with $\|X\| = \|Y\| = 1$, $\Delta(X) = \delta \in [0,1)$ and $\Delta(Y) = \delta' \in [0,1]$, and every unitary (orthogonal) $U \in \mathcal{M}$,*

$$|\,\mathrm{Tr}(XUY)| - \mathrm{Tr}(XY) \leq \frac{N\,\delta^2\delta'^2}{(1+\delta)(1+\delta')},$$

and the bound is attained. Equivalently, $\sup\{|\,\mathrm{Tr}(XZ)| - \mathrm{Tr}(X|Z|) : \|X\| = \|Z\| = 1,\ \Delta(X) = \delta,\ \Delta(|Z|) = \delta'\}$ has this value.

Proof. Let $\ell = (1-\delta)/(1+\delta)$ and $\ell' = (1-\delta')/(1+\delta')$ be the smallest eigenvalues. It suffices to prove the bound in the complex algebra, since the maximisers below are real. By the remark above, $\sup_U z = F(X,Y) := \|YX\|_1 - \mathrm{Tr}(XY)$, and the polar unitary lies in $\mathcal{M}$. For a fixed eigenbasis of X, F is a convex function of the eigenvalues of X (a norm of a linear function of them minus a linear function). The admissible eigenvalues form the union of the faces $\{x_i = 1,\ x_j = \ell\}$ of the cube $[\ell,1]^K$, $K = \sum_c k_c$, and a convex function on a face attains its maximum at a vertex; so $F(X,Y) \le F(X',Y)$ for some $X' = \ell + (1-\ell)P$ with a projection $P \in \mathcal{M}$, $P \neq 0,1$, and likewise $F(X',Y) \le F(X',Y')$ with $Y' = \ell' + (1-\ell')Q$. By Lemma 2.2, applied in every block, X' and Y' are direct sums over subspaces of dimension 1 or 2, and so are $\|Y'X'\|_1$ and $\mathrm{Tr}(X'Y')$. One-dimensional summands contribute 0. On a two-dimensional one, $X' = \mathrm{diag}(1,\ell)$ and Y' has the eigenvalues $1, \ell'$, so by Proposition 4.2 it contributes at most $(1-\ell)^2(1-\ell')^2/(4(1+\ell)(1+\ell')) = \delta^2\delta'^2/((1+\delta)(1+\delta'))$. A block of size k_c holds at most $\lfloor k_c/2 \rfloor$ two-dimensional summands. Equality: N blocks $\mathrm{diag}(1,\ell)$ and Y with spectrum $\{1,\ell'\}$ at the angle of Proposition 4.2, $X = Y = 1$ on the remaining one-dimensional blocks, and the polar unitary, which makes all block traces positive. □

Remark 4.4. (i) With the contrast of $|Z|$ free, the supremum $N\delta^2/(2(1+\delta))$ is reached at $\delta' = 1$, i.e. for singular Z. (ii) The bound is additive over the blocks: it counts the mutually orthogonal two-dimensional subspaces that fit into the blocks, and it does not depend on the sizes of the blocks otherwise. (iii) The normalised version is Theorem 3.1: $\sup_Z |\mathrm{Tr}(XZ)|/\mathrm{Tr}(X|Z|) = \kappa(X)$, attained by a rank-one $|Z|$, i.e. at $\Delta(|Z|) = 1$. The unnormalised supremum vanishes to fourth order at $\delta = \delta' = 0$ because the violation needs both operators to be far from scalar. (iv) Figures 1 and 2 compare the bound with random sampling and with numerical optimisation. Sampling, the method of [1], stays far below the bound as soon as the blocks are larger than 2×2, while the optimisation over both contrasts reproduces it.

5 Young, Bikchentaev–Abed and the canonical trace

Proposition 5.1. *$\varphi_W(|AB^*|) \le h \cdot \frac{1}{2}\big(\varphi_W(|A|^2) + \varphi_W(|B|^2)\big)$ for all A, B, with equality for $A = B = e_1e_2^*$. Hence $V_n = h - 1 = 2\delta/(1-\delta)$ for item 8 in every dimension.*

Proof. $\varphi_W(|AB^*|) \le a\|AB^*\|_1 \le a\|A\|_2\|B\|_2$ and $\varphi_W(|A|^2) = \mathrm{Tr}(WA^*A) \ge b\|A\|_2^2$. For $A = B = e_1e_2^*$, $|AB^*| = e_1e_1^*$ and $|A|^2 = e_2e_2^*$, so $L = a$ and $R = b$. □

Remark 5.2. For the published form with $A, B \ge 0$ an independent numerical search ($n = 2, 3$) gives only $V \approx \kappa - 1$ (0.1546 at $\delta = 0.5$ and 1.294 at $\delta = 0.9$, against $\kappa - 1 = 0.1547$ and 1.294). The linear failure of item 8 is a feature of the extension to arbitrary operators.

Proposition 5.3. *Let $\Delta(W) = \delta$ and let the moduli of A and B have contrast δ'. For $n = 2$ (and, by direct sums, for every even n) the extended Young inequality has*

$$V_n(\delta,\delta') \ \ge\ \frac{h + \ell'^2}{1 + h\ell'^2} - 1 = \frac{4\delta\delta'}{1 + \delta'^2 - 2\delta\delta'}, \qquad \ell' = \frac{1-\delta'}{1+\delta'},$$

attained by $A = B = UT$ with $T = \mathrm{diag}(\ell', 1)$ in the eigenbasis of $W = \mathrm{diag}(a,b)$ and U the swap of the two basis vectors. For $\delta' = 1$ this is Proposition 5.1.

Proof. For $A = B$, $|AB^*| = |AA^*| = AA^* = UT^2U^* = \mathrm{diag}(1, \ell'^2)$ and $|A|^2 = T^2 = \mathrm{diag}(\ell'^2, 1)$, so $L = a + b\ell'^2$ and $R = a\ell'^2 + b$, and $L/R = (h + \ell'^2)/(h\ell'^2 + 1)$. Inserting $h = \frac{1+\delta}{1-\delta}$ and ℓ' gives the second form. □

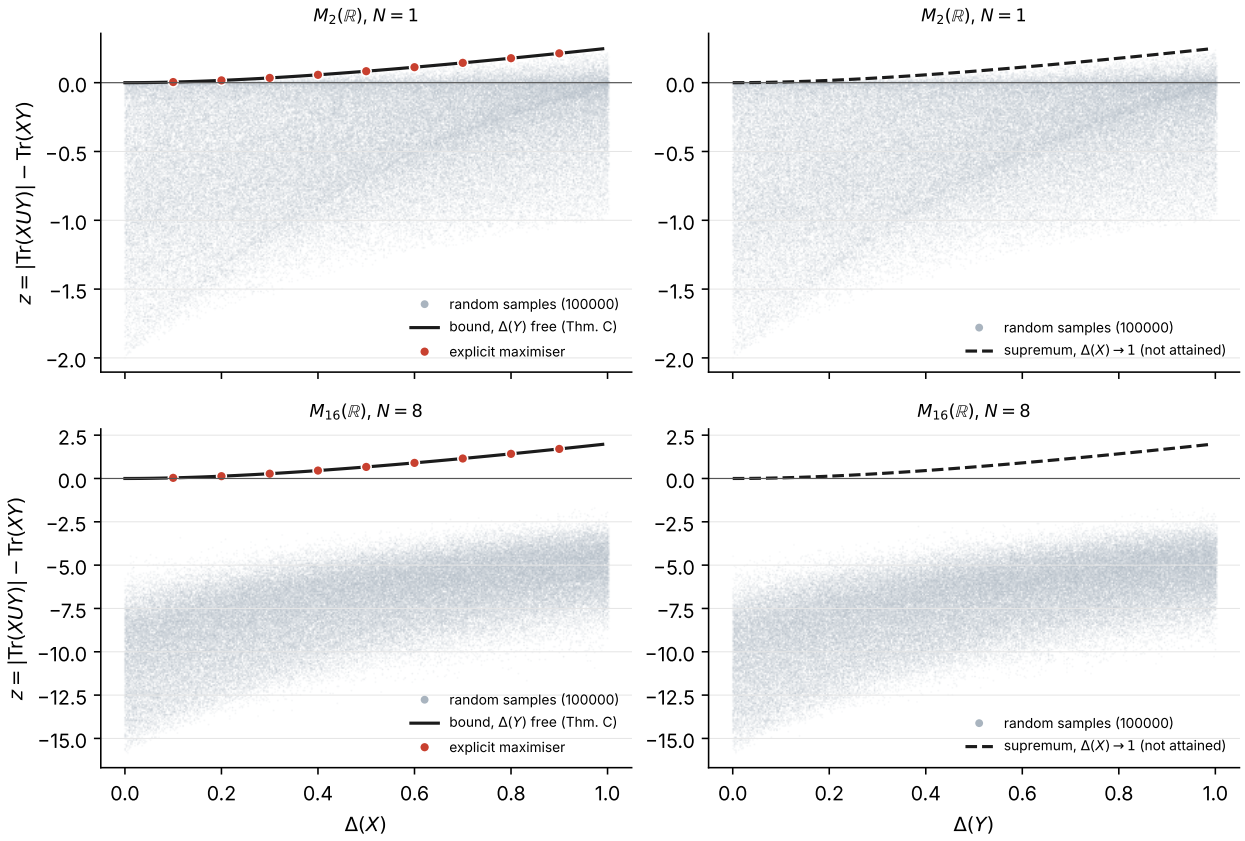


Figure 1: The unnormalised Gardner violation $z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY)$ for $\|X\| = \|Y\| = 1$ in $M_2(\mathbb{R})$ (top, $N = 1$) and $M_{16}(\mathbb{R})$ (bottom, $N = 8$), against $\Delta(X)$ (left) and $\Delta(Y)$ (right). Grey: 100000 random samples with Haar-orthogonal eigenbases and U and uniformly drawn contrasts, as in [1, Fig. 1]. Solid: the bound of Theorem 4.3 with $\Delta(Y)$ free, $N\delta^2/(2(1+\delta))$; dots: z at the explicit maximisers of its proof, which agree with it to 10^{-15}. Dashed: the supremum along $\Delta(Y)$, approached as $\Delta(X) \to 1$. In M_{16} no sample violates Gardner's inequality ($z > 0$), although violations up to almost 2 exist: the maximisers, two-level spectra on aligned pairs of eigenvectors, form a set of negligible Haar measure. Computed with plain NumPy by `scripts/trace_envelopes_figures.py`.

The certified search over all A, B with this contrast reproduces the bound to four digits on M_2 (Figure 16); we conjecture equality for even n. Unlike Gardner's inequality, the extended Young inequality fails *linearly* in each contrast, $V \sim 4\delta\delta'$.

Proposition 5.4. *For $n = 2$, item 9 has the envelope $V_2 = s/(1+s)$, $s = \sin(\psi/2)$. Thus $V_2 \sim \delta/2$ as $\delta \to 0$ and $V_2 \to \sqrt{2} - 1$ as $\delta \to 1$.*

Proof. If P or Q is 0 or 1, or $P = Q$, then $L = R$. Otherwise $P = uu^*$, $Q = vv^*$ with unit vectors; choose phases so that $u = cv + \sigma v_\perp$ with $c = \cos\theta$, $\sigma = \sin\theta$, $0 < \theta < \pi/2$, and a unit vector $v_\perp \perp v$. Normalise $\operatorname{Tr} W = 2$, so that W has the eigenvalues $1 \pm \delta$, and put $\beta = v^*Wv$, $\gamma = v_\perp^* W v_\perp = 2 - \beta$, $t = \operatorname{Re}(v^*Wv_\perp)$ and $\rho = |v^*Wv_\perp|$; then $\beta\gamma - \rho^2 = \det W = 1 - \delta^2$. Since $PQ + QP = c(uv^* + vu^*)$ and $QPQ = c^2 Q$,

$$L - R = 2c\operatorname{Re}(v^*Wu) - 2c^2\beta = 2c\sigma t, \qquad N = u^*Wu + \beta = \beta(1+c^2) + \sigma^2\gamma + 2c\sigma t.$$

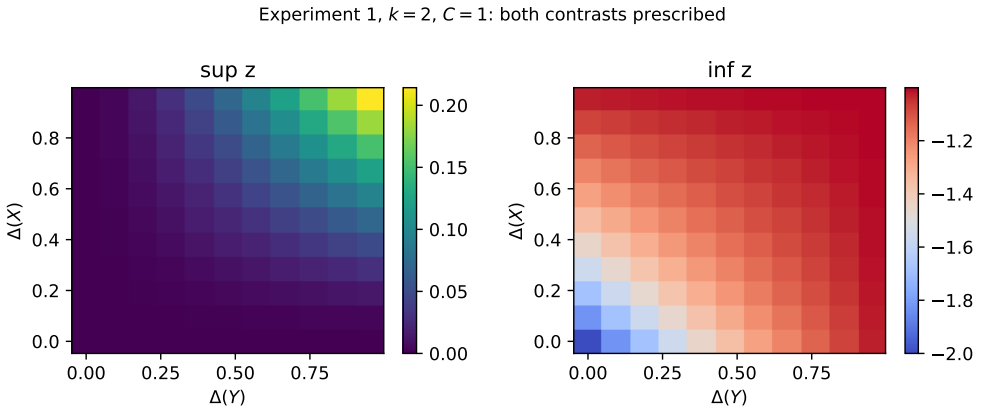


Figure 2: $M_2(\mathbb{R})$, both contrasts prescribed: the supremum (left) and infimum (right) of $z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY)$ found by constrained optimisation on a grid of $\Delta(Y)$ (horizontal) and $\Delta(X)$ (vertical). The left panel agrees with $\delta^2\delta'^2/((1+\delta)(1+\delta'))$ (Theorem 4.3) to 5×10^{-10} at every grid point: a violation needs both operators to be far from scalar.

With $D = \beta(1+c^2) + \sigma^2\gamma > 0$ the quotient $2c\sigma t/(D + 2c\sigma t)$ increases with $t \le \rho$, so it is at most $r/(1+r)$ with

$$r = \frac{2c\sigma\rho}{D} = \frac{\rho \sin 2\theta}{\beta + 1 + (\beta - 1)\cos 2\theta} \le \frac{\rho}{\sqrt{(\beta+1)^2 - (\beta-1)^2}} = \frac{\rho}{2\sqrt{\beta}},$$

using $\gamma = 2 - \beta$ and $\max_x \sin x/(p + q\cos x) = (p^2 - q^2)^{-1/2}$ for $p > |q|$. Finally $r^2 \le \rho^2/(4\beta) = \frac{1}{4}\big(2 - \beta - (1-\delta^2)/\beta\big)$, which is maximal at $\beta = \sqrt{1-\delta^2} = \cos\psi$ with the value $\frac{1}{2}(1 - \cos\psi) = \sin^2(\psi/2)$. All steps are equalities for v with $v^*Wv = \cos\psi$, a phase of $v_\perp$ with $v^*Wv_\perp = \rho > 0$, and the optimal θ, and $r \mapsto r/(1+r)$ is increasing. □

Proposition 5.5. *Let* $\operatorname{Tr} W = n$. *For item 11,* $V(W) = 1 - \mathrm{GM}/\mathrm{AM}$ *of the eigenvalues of* W, *attained at* $A = W^{-1}$. *Over the densities with contrast* δ *the maximum is attained at a two-level spectrum:*

$$V_n(\delta) = 1 - \min_{1 \le m < n} \frac{n\,\ell^{(n-m)/n}}{m + (n-m)\ell}, \qquad \ell = \frac{1}{h} = \frac{1-\delta}{1+\delta}.$$

In particular $V_2 = 1 - \sqrt{1-\delta^2}$, *and* $V_n \uparrow 1 - 1/S(h)$ *as* $n \to \infty$, *where* $S(h) = \frac{(h-1)h^{1/(h-1)}}{e \log h}$ *is the Specht ratio.*

Proof. By the AM–GM inequality for the eigenvalues of $W^{1/2}AW^{1/2}$, $\varphi_W(A)/n \ge (\det W \det A)^{1/n}$, with equality for $A = W^{-1}$, so $\inf_{A>0} \varphi_W(A)/(n(\det A)^{1/n}) = (\det W)^{1/n}$ and $V(W) = 1 - (\det W)^{1/n}$, which for $\operatorname{Tr} W = n$ is $1 - \mathrm{GM}/\mathrm{AM}$. In the coordinates $x_i = \log w_i$, $\mathrm{AM}/\mathrm{GM} = \frac{1}{n}\sum_i e^{x_i - \bar{x}}$ is convex, so on the box of spectra in $[b, a]$ it is maximal at a vertex, a two-level spectrum with m eigenvalues a and $n - m$ eigenvalues b; this gives the formula. Its limit is Specht's theorem [31]: $\sup \mathrm{AM}/\mathrm{GM}$ over all finite families in $[b, a]$ is $S(h)$. □

Proposition 5.6. *Let* $\operatorname{Tr} W = n$. *For item 24,* $V(W) = w_{\min}^{-1/2} - 1$, *and*

$$V_n(\delta) = \sqrt{\frac{\ell + n - 1}{n\ell}} - 1, \qquad \ell = \frac{1-\delta}{1+\delta};$$

$V_2 = (1-\delta)^{-1/2} - 1 \sim \delta/2$, $V_n \sim \frac{n-1}{n}\delta$, *and* $V_n \uparrow \sqrt{h} - 1$ *as* $n \to \infty$.

Proof. For $A \geq 0$, $\varphi(A^2) = \mathrm{Tr}(A^{1/2} W A^{1/2} A) \leq \|A\| \varphi(A) \leq \mathrm{Tr}(A) \varphi(A) \leq \varphi(A)^2 / w_{\min}$, with equality for the projection onto an eigenvector for $w_{\min}$. With $\mathrm{Tr}\, W = n$ and the extreme eigenvalues in the ratio h, $w_{\min} = nb / \sum_i w_i \geq nb/(b + (n-1)a) = n\ell/(\ell + n - 1)$, with equality for the spectrum $\{a, \ldots, a, b\}$. For $n = 2$, $(\ell + 1)/(2\ell) = 1/(1 - \delta)$. □

Remark 5.7. Items 11 and 24 characterise the canonical trace among the functionals with $\varphi(1) = n$, so they are violated also by central densities on a direct sum (Section 8), and their envelopes grow with n, unlike all others with a closed form.

6 Inequalities for two projections

Theorem 6.1. *Let $W > 0$ in M_n, $n \geq 2$, with $\Delta(W) = \delta$, and let P, Q be projections. With N as in Table 1:*

(17) $V_n = \kappa^2 - 1$ *for* $\varphi(PQP) \leq \varphi(Q)$;

(18) $V_n = \kappa - 1$ *for* $\varphi(|P - Q|) \leq \varphi(P + Q)$;

(19) $V_n = \delta$ *for* $\varphi(P + Q) \leq \varphi(1 + P \wedge Q)$ *if n is even, and* $V_n = (n-1)\delta/(n - \delta)$ *if n is odd;*

(20) for $\varphi(e^{QPQ}) \leq \varphi(e^P)$ *and even n,* $V_n = \max_{0<g<\pi} \lambda(\delta, g)$, *where $\lambda(\delta, g)$ is the larger root of the quadratic* (2) *below, and* $V_n = \frac{(e-1)^2}{e(e+1)} \delta^2 + O(\delta^3)$;

(21) $V_n = \frac{1}{2} \tan^2(\psi/2)$ *for* $\varphi(\mathrm{Re}\, PQ) \geq 0$;

(22) $V_n = \kappa - 1$ *for* $\varphi(\mathrm{Re}\, PQ) \leq \varphi(|PQ|)$;

(23) $V_n = \delta/(1 - \delta)$ *for* $\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$.

The values are attained in 17, 18, 20–22 by rank-one P, Q, and approached in 19 and 23 as $P \neq Q$ tend to the same eigenvector of W. For odd n, V_n in item 20 lies between the M_2 value and the same maximum with $(1 - \delta)/(n - 1)$ added to β in (2).

Proof. Step 1: reduction to rank-one projections on M_2. By Lemma 2.2, $\mathbb{C}^n = \bigoplus_j H_j$ with $\dim H_j \in \{1, 2\}$, every H_j invariant under P and Q. Every operator in L, R and N (namely PQP, Q, $|P - Q|$, $P + Q$, $1 + P \wedge Q$, e^{QPQ}, e^P, $\mathrm{Re}\, PQ$, $|PQ|$, $P \vee Q$) is then block diagonal, with the block obtained from P_j, Q_j; for the lattice operations because $\mathrm{ran}\, P \cap \mathrm{ran}\, Q = \bigoplus_j (\mathrm{ran}\, P_j \cap \mathrm{ran}\, Q_j)$, the projections onto the H_j commuting with P and Q. Hence $L = \sum_j L_j$, $R = \sum_j R_j$ and $N = \sum_j N_j$, where L_j is the same expression for P_j, Q_j and the functional $\mathrm{Tr}(W_j \cdot)$ on H_j, with the compression W_j of W to H_j; its spectrum lies in $[b, a]$, so $\Delta(W_j) \leq \delta$. On a one-dimensional H_j, $P_j, Q_j \in \{0, 1\}$ commute and $L_j \leq R_j$ in all seven items (e.g. $\max(p, q) \leq p + q$, $p + q \leq 1 + pq$, $e^{pq} \leq e^p$). On a two-dimensional H_j, $P_j \neq Q_j$ are rank-one projections. Since $N_j \geq 0$, and $L_j \leq R_j$ wherever $N_j = 0$,

$$\frac{L - R}{N} \leq \max_j \frac{L_j - R_j}{N_j},$$

and the quotients do not change when W_j is scaled. It therefore suffices to bound $(L - R)/N$ for $W = \frac{1}{2}(1 + \tau\sigma_3)$, $0 \leq \tau \leq \delta$, and $P = \frac{1}{2}(1 + \mu \cdot \sigma)$, $Q = \frac{1}{2}(1 + \nu \cdot \sigma)$ with unit Bloch vectors $\mu \neq \nu$ in $\mathbb{R}^3$ ($\sigma = (\sigma_1, \sigma_2, \sigma_3)$ the Pauli matrices). For fixed μ, ν the quotient is a linear-fractional function of τ (all three quantities are affine in W) with positive denominator, hence monotone on $[-\delta, \delta]$; conjugation by σ_1 maps τ to $-\tau$ and (μ, ν) to reflected vectors. So the bound for $\tau = \delta$ covers every $\tau \leq \delta$, and the M_2 envelope bounds the envelope in every dimension. In 17, 18 and 21–23, N vanishes on blocks

where $P = Q = 0$, so the values below, attained (or approached) on M_2, are attained (approached) on M_n with P, Q supported on the span of e_1, e_2. In 19 and 20, N contains $\varphi(1)$, which we treat at the end.

Step 2: formulas on M_2. Write $\varphi(x_0 + x \cdot \sigma) = x_0 + \delta x_3$, $\mu \cdot \nu = \cos 2\theta$ $(0 < \theta \le \pi/2)$ and $c = \cos\theta = |\langle u, v\rangle|$ for unit vectors u, v with $P = uu^*$, $Q = vv^*$. Then $PQP = c^2 P$, $(P - Q)^2 = \sin^2\theta \cdot 1$, $P \wedge Q = 0$, $P \vee Q = 1$, $\operatorname{Re} PQ = \frac{1}{4}(1 + \mu \cdot \nu + (\mu + \nu) \cdot \sigma)$, $|PQ| = (QPQ)^{1/2} = cQ$, $e^{QPQ} = 1 + (e^{c^2} - 1)Q$, $e^P = 1 + (e - 1)P$, and $|\mu + \nu| = 2\cos\theta$. We use the following elementary fact: *for $\beta > |z|$, $\max_{|\xi|=1}(\alpha + y \cdot \xi)/(\beta + z \cdot \xi)$ is the unique zero λ of $\alpha - \lambda\beta + |y - \lambda z|$*, since $(\alpha + y \cdot \xi) - \lambda(\beta + z \cdot \xi) \le \alpha - \lambda\beta + |y - \lambda z|$ with equality for $\xi \parallel y - \lambda z$, and the right-hand side decreases strictly in λ.

(17) $L/R = (1 + \mu \cdot \nu)(1 + \delta\mu_3)/(2(1 + \delta\nu_3))$. For fixed μ, the fact with $\alpha = \beta = 1$, $y = \mu$, $z = \delta e_3$ gives $\max_\nu (1 + \mu \cdot \nu)/(1 + \delta\nu_3) = 2(1 - \delta\mu_3)/(1 - \delta^2)$, so $L/R \le (1 - \delta^2\mu_3^2)/(1 - \delta^2) \le \kappa^2$, with equality for $\mu_3 = 0$ and $\nu = (\mu - \lambda\delta e_3)/(\lambda - 1)$, $\lambda = 2/(1 - \delta^2)$.

(18) $L/R = \sin\theta/(1 + \frac{\delta}{2}(\mu_3 + \nu_3)) \le \sin\theta/(1 - \delta\cos\theta) \le \kappa$, the maximum being at $\cos\theta = \delta$, $\mu + \nu \parallel -e_3$. (Alternatively, $\varphi(|P - Q|) \le \kappa\varphi(P + Q)$ is the triangle inequality of Theorem 3.1 with $X = P$, $Y = -Q$.)

(19) $(L - R)/N = \varphi(P + Q) - 1 = \frac{\delta}{2}(\mu_3 + \nu_3) < \delta$, approached as $\mu, \nu \to e_3$.

(20) For fixed $g = 2\theta$, the pair (μ_3, ν_3) ranges over the image of the unit sphere under $\xi \mapsto (\mu \cdot \xi, \nu \cdot \xi)$, a filled ellipse, and the linear-fractional quotient is maximal on its boundary, $\mu_3 = \cos x$, $\nu_3 = \cos(x + g)$. With $K = e - 1$ and $E = e^{\cos^2(g/2)} - 1$,

$$\frac{L - R}{N} = \frac{\alpha + y \cdot \xi_x}{\beta + z \cdot \xi_x}, \quad \alpha = \tfrac{E-K}{2},\ y = \tfrac{\delta}{2}(E\cos g - K,\ -E\sin g),\ \beta = 1 + \tfrac{K}{2},\ z = (\tfrac{\delta K}{2}, 0),$$

$\xi_x = (\cos x, \sin x)$. By the fact, its maximum $\lambda(\delta, g)$ is the larger root of

$$(\beta^2 - |z|^2)\lambda^2 - 2(\alpha\beta - y \cdot z)\lambda + \alpha^2 - |y|^2 = 0. \tag{2}$$

For small δ the maximiser has $g = 2K\delta/e + O(\delta^2)$, $x = \pi/2$, and $L - R = -\frac{e}{8}g^2 + \frac{K\delta}{2}g + O(\delta^3)$, which gives $V = K^2\delta^2/(e(e + 1)) + O(\delta^3)$.

(21) With $s = \mu_3 + \nu_3 \ge -2\cos\theta$, $(L - R)/N = -(2\cos^2\theta + \delta s)/(4(1 + \frac{\delta}{2}s))$, which decreases in s; so $(L - R)/N \le (\delta C - C^2)/(2(1 - \delta C))$, $C = \cos\theta$. The maximum over C is at $\delta C^2 - 2C + \delta = 0$, $C = \tan(\psi/2)$, with the value $C^2/2 = \frac{1}{2}\tan^2(\psi/2)$.

(22) $\varphi(\operatorname{Re} PQ) = \operatorname{Re}\varphi(PQ) \le |\varphi(PQ)| \le \kappa\varphi(|PQ|)$ by Theorem 3.1, in every dimension. Equality: $v = \sqrt{b/(a + b)}\, e_1 + \sqrt{a/(a + b)}\, e_2$, $u = Wv/\|Wv\|$; then $PQ = \langle u, v\rangle uv^*$ has the polar factor of the equality case of Theorem 3.1.

(23) $L/R = 1/(1 + \frac{\delta}{2}(\mu_3 + \nu_3)) < 1/(1 - \delta)$, approached as $\mu, \nu \to -e_3$.

Step 3: items 19 and 20 in dimension n. For even n take W with the eigenvalues a and b, each $n/2$ times, and $n/2$ copies of the M_2 extremiser, one in each block spanned by an eigenvector for a and one for b; this attains (approaches) the M_2 value, which is an upper bound by Step 1. For odd n at least one block of Lemma 2.2 is one-dimensional, so there are $k' \le k = (n - 1)/2$ two-dimensional blocks. In item 19, $N \ge \varphi(1) = \operatorname{Tr} W$; by Step 2 a two-dimensional block contributes $L_j - R_j = \operatorname{Tr} W_j \cdot \frac{\Delta(W_j)}{2}(\mu_3 + \nu_3) < \lambda_1(W_j) - \lambda_2(W_j)$, and a one-dimensional one $L_j - R_j \le 0$. By Ky Fan's maximum principle [3, Sect. III.1], $\sum_j \lambda_1(W_j)$ is at most the sum of the k' largest and $\sum_j \lambda_2(W_j)$ at least the sum of the k' smallest eigenvalues of W. The resulting bound for $(L - R)/N$ is a linear-fractional function of the spectrum of W and is maximal for $k' = k$ and the spectrum $a^{(k)}, b^{(k+1)}$; hence $(L - R)/N < k(a - b)/(k(a + b) + b) = (n - 1)\delta/(n - \delta)$, which is approached for this spectrum with $P = 1$, $Q = 0$ on the remaining eigenvector (a block with $L_j = R_j$). In item 20 the same configuration, with $P = Q = 0$ on the remaining eigenvector, adds b/k per block to N, i.e. $(1 - \delta)/(n - 1)$ to β (normalisation $\operatorname{Tr} W_j = 1$); this gives the lower bound, and the M_2 value the upper bound. $\square$

7 Lower bounds

Proposition 7.1. *For every $n \geq 2$:*

(a) $V_n \geq \kappa - 1$ *for item 6 (approached);*

(b) $V_n \geq \kappa^2 - 1 = \delta^2/(1-\delta^2)$ *for item 7 (approached);*

(c) $V_n \geq m(\delta) := \max_{0<y<\delta} \dfrac{y^2(\delta-y)^2}{(1-2\delta y+y^2)(1-y^2)} = \dfrac{\delta^4}{16} + O(\delta^5)$ *for item 10 (attained);*

(d) $V_n \geq \delta^2/36 + O(\delta^3)$ *for item 14, and more generally* $(1-\frac{2}{p})^2\delta^2/4 + O(\delta^3)$ *for Hölder's inequality* $|\varphi(PQ)| \leq \varphi(P)^{1/p}\varphi(Q)^{1/q}$.

By Lemma 2.3, $V_n \leq h - 1$ *in (a)–(c).*

Proof. All constructions live on $\operatorname{span}\{e_1, e_2\}$ (extended by 0, or by a small multiple of the identity where positive definiteness is required); let $e_\pm = (e_1 \pm e_2)/\sqrt{2}$.

(a) For invertible A and a rank-one $B = vv^*$, $\|v\| = 1$, we have $A^{-1}\#vv^* = (v^*Av)^{-1/2}vv^*$, hence $(A^{-1}\#B)^{1/2} = (v^*Av)^{-1/4}vv^*$ and $A\natural B = (v^*Av)^{1/2}vv^*$. With $v = e_+$ and $A = xx^* + \varepsilon$,

$$\frac{\varphi(A\natural B)^2}{\varphi(A)\varphi(B)} = \frac{(v^*Av)(v^*Wv)}{\operatorname{Tr}(WA)} \xrightarrow[\varepsilon\to 0]{} \frac{|v^*x|^2\, v^*Wv}{x^*Wx},$$

which for $x = W^{-1}v$ equals $(v^*W^{-1}v)(v^*Wv) = \frac{1}{4}(a^{-1}+b^{-1})(a+b) = \kappa^2$. By continuity the quotient L/R approaches κ for positive definite A, $B = vv^* + \varepsilon'$.

(b) In the basis $f_1 = e_+$, $f_2 = e_-$ the density has the entries $W_{11} = W_{22} = \frac{a+b}{2}$, $W_{12} = \frac{a-b}{2}$. Take $B = \eta f_1 f_1^* + f_2 f_2^*$ and $A = uu^*$ with $u = f_1 + s\eta f_2$. Then $B^{1/2}AB^{1/2} = ww^*$ with $w = \sqrt{\eta} f_1 + s\eta f_2$, so $L = \|w\|^2\, w^*Ww = \eta^2 W_{11} + O(\eta^{5/2})$, and $Bu = \eta(f_1 + sf_2)$, so $R = \|u\|^2\eta^2(W_{11} + 2sW_{12} + s^2W_{22})$. As $\eta \to 0$, $L/R \to W_{11}/(W_{11} + 2sW_{12} + s^2W_{22})$, which at $s = -W_{12}/W_{22}$ equals $W_{11}W_{22}/(W_{11}W_{22} - W_{12}^2) = (a+b)^2/(4ab) = \kappa^2$.

(c) Normalise $a + b = 2$, so $W = \operatorname{diag}(1+\delta, 1-\delta)$, and take $A = uu^*$, $B = A + tvv^*$, $t \geq 0$, with $u = (\cos\alpha, \sin\alpha)$, $v = (-\cos\alpha, \sin\alpha)$ and $y = -\cos 2\alpha$. Then $u^*v = y$, $v^*Wu = y - \delta$ and $u^*Wu = v^*Wv = 1 - \delta y$, so

$$L - R = \varphi(A^2) - \varphi(B^2) = -2t\operatorname{Re}\big((u^*v)(v^*Wu)\big) - t^2 v^*Wv = 2ty(\delta - y) - t^2(1 - \delta y).$$

For $0 < y < \delta$ the best $t = y(\delta - y)/(1 - \delta y) > 0$ gives $L - R = G := y^2(\delta - y)^2/(1 - \delta y)$ and $(L-R)/R = G/(1 - \delta y - G) = y^2(\delta - y)^2/((1 - 2\delta y + y^2)(1 - y^2))$. Maximising over y gives $m(\delta)$; the leading order comes from $y = \delta/2$.

(d) Take $W = \frac{1}{2}(1 + \delta D)$, $D = \operatorname{diag}(1, -1)$, and real unit vectors u, v at the angles $m \pm \varepsilon/2$ to e_1, $P = uu^*$, $Q = vv^*$. Then $\varphi(PQ) = \frac{1}{2}(\cos^2\varepsilon + \delta\cos\varepsilon\cos 2m)$ and $\varphi(P)^{1/p}\varphi(Q)^{1/q} = \frac{1}{2}\big(1 + \delta\cos 2m + \delta(\frac{1}{q} - \frac{1}{p})\varepsilon\sin 2m\big) + O(\delta^2\varepsilon^2)$, so $L - R = \frac{1}{2}\big(-\varepsilon^2 - \delta(1 - \frac{2}{p})\varepsilon\sin 2m\big) + O(\delta^3)$ for $\varepsilon = O(\delta)$. The choice $m = \pi/4$, $\varepsilon = -\frac{\delta}{2}(1 - \frac{2}{p})$ gives $L - R = \frac{1}{8}(1 - \frac{2}{p})^2\delta^2 + O(\delta^3)$ and $R = \frac{1}{2} + O(\delta)$. □

Proposition 7.2. *For item 25 ($\varphi(A^4) \leq \varphi(AB^2A)$ for $0 \leq A \leq B$), rank-one A gives no violation, and an explicit construction with A of rank two and $B - A$ of rank one gives*

$$V_n \geq m_{25}(\delta) = \frac{\delta^8}{4096}\big(1 + O(\delta^2)\big) \qquad \text{in every dimension } n \geq 2.$$

Proof. If $A = \alpha uu^*$ has rank one and $A \le B$, then $u^*Bu \ge \alpha$ and $AB^2A = \alpha^2\|Bu\|^2uu^* \ge \alpha^2(u^*Bu)^2uu^* \ge A^4$, so every positive φ satisfies the inequality. For the construction take W with $\operatorname{Tr} W = 1$ and eigenvalues $(1 \pm \delta)/2$, and, in an orthonormal basis at the angle ϑ to its eigenbasis, $A = \operatorname{diag}(1, s)$, $0 < s < 1$, and $B = A + tvv^*$ with $v = (\cos\phi, \sin\phi)$; in this basis W has the entries $w_{11,22} = \frac{1}{2}(1 \pm \delta\cos 2\vartheta)$ and $w_{12} = \frac{\delta}{2}\sin 2\vartheta$. Then $R - L = tT_1 + t^2T_2$ with $T_1 = \varphi(A^2P_vA + AP_vA^2)$, $T_2 = \varphi(AP_vA)$, $P_v = vv^*$, and whenever $T_1 < 0$ the best $t = -T_1/(2T_2)$ gives $(L-R)/R = G/(L-G)$, $G = T_1^2/(4T_2)$. Let $m_{25}(\delta)$ be the maximum over (ϑ, s, ϕ); it is attained, and by embedding $\mathbb{C}^2 \subset \mathbb{C}^n$ (with A, B extended by 0) it is a lower bound in every dimension. For the leading order put $\vartheta = \pi/4$, $\sqrt{s} = \sigma\delta$ and $\tan\phi = -\varrho s^{-3/2}$. Then $T_1 = \cos^2\phi\,(1 + \varrho^2 - (1+s)\varrho/\sigma)$, $T_2 = \cos^2\phi\,(\frac{1}{2} + \frac{\varrho^2}{2s} - \frac{\varrho}{\sigma})$ and $\cos^2\phi = s^3/(s^3 + \varrho^2)$, so that

$$\frac{L-R}{R} = \big(\sigma^2(\varsigma - \varsigma^2) - \sigma^4\big)^2\delta^8 + O(\delta^{10}), \qquad \varsigma = \sigma/\varrho,$$

which is maximal for $\varsigma = \frac{1}{2}$, $\sigma^2 = \frac{1}{8}$, with the value $\delta^8/4096$. □

The numerical maximisation gives $m_{25}(0.1)/(0.1^8/4096) = 1.020$. The same scaling for a general exponent $r \ge 0$ in $\varphi(A^{2r+2}) \le \varphi(A^rB^2A^r)$ gives the order $\delta^{4(r+1)}$; for $r = 0$ the supremum of the leading coefficient is $\frac{1}{16}$, that of item 10.

Remark 7.3. With $N = R$, item 13 would have $V = \infty$ for every $\delta > 0$: for rank-one $A = uu^*$, $B = vv^*$ with overlap $c = |u^*v| \to 0$, $L \approx 2c\,|u^*Wu_\perp|$ while $R = 2c^2u^*Wu$. This is why we normalise by $\varphi(A + B)$.

8 Direct sums: only the contrast inside the blocks matters

Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}$, with the trace $\operatorname{Tr} = \sum_c \operatorname{Tr}_c$ and $\varphi_W = \operatorname{Tr}(W\,\cdot\,)$, $W = \bigoplus_c W_c \ge 0$. The traces on $\mathcal{M}$ are the functionals $\sum_c t_c \operatorname{Tr}_c$, $t_c \ge 0$; so φ_W is a trace iff W is central, whatever its global contrast. The global contrast $\Delta(W)$ thus says nothing about traciality, and the following theorem shows that it says nothing about the size of the violation either.

Theorem 8.1. *Let $W = \bigoplus_c W_c > 0$ in $\mathcal{M}$, and let $V_{\mathcal{M}}(W)$ and $V_{M_{k_c}}(W_c)$ be the suprema of the normalised violation of one inequality of Table 1, other than 11, 16 and 24, for φ_W on $\mathcal{M}$ and for φ_{W_c} on M_{k_c}. Then*

$$V_{\mathcal{M}}(W) \le \max_c V_{M_{k_c}}(W_c),$$

with equality for items 1–10, 12–14, 17, 18, 21–23 and 25. In particular, if every W_c is scalar, no inequality of Table 1 except 11 and 24 is violated, however different the scalars are; and the envelope on $\mathcal{M}$ along the intra-block contrast $\max_c \Delta(W_c) = \delta$ is at most the largest envelope $V_{k_c}(\delta)$ of the blocks.

Proof. All variables of the items are in $\mathcal{M}$, hence block diagonal, and every functional calculus acts blockwise. Let L_c, R_c, N_c be the expressions of the item for the block variables and φ_{W_c}. For all items except 11, 16 and 24, $L \le \sum_c L_c$ (with equality unless L contains an absolute value of a functional, as in 1, 4 and 14), $N = \sum_c N_c$, and $R = \sum_c R_c$, except in items 6 and 14, where $R = \sqrt{\sum_c \varphi_c(A_c)\sum_c \varphi_c(B_c)} \ge \sum_c R_c$ by the Cauchy–Schwarz inequality and $R \ge \sum_c R_c$ by Hölder's inequality. Since $W > 0$, $N_c = 0$ forces the variables of block c to vanish where they enter L_c, so $L_c \le R_c$ there. Hence $(L - R)/N \le \sum_c(L_c - R_c)/\sum_c N_c \le \max_c(L_c - R_c)/N_c$, the maximum over the blocks with $N_c > 0$ (for $N = R$ in items 6 and 14 use $L/R \le \sum_c L_c/\sum_c R_c$ instead). This proves the inequality. For the equality, in the listed items N is additive and the variables may vanish on all blocks but one without making N vanish; restricting a near-optimal configuration of one block and

setting the variables to zero on the other blocks reproduces its quotient (in 19, 20 and 15, N contains $\varphi(1)$ and the other blocks dilute the violation). If every W_c is scalar, every $V_{M_{k_c}}(W_c)$ vanishes, since φ_{W_c} is then a trace on M_{k_c}; for item 16 the same conclusion holds because the Peierls–Bogoliubov inequality holds for every trace on $\mathcal{M}$. □

Remark 8.2. The theorem is a statement about the characterisations, not about the inequalities: a non-central density is detected by every characterisation of traces, but only through the variation *inside* the blocks. Items 11 and 24 characterise the *canonical* trace Tr among the functionals with $\varphi(1) = \dim$, and central W violate them; for a fixed density their violations are $1 - \mathrm{GM/AM}$ and $w_{\min}^{-1/2} - 1$ of the eigenvalues of W normalised to $\operatorname{Tr} W = \sum_c k_c$ (Propositions 5.5 and 5.6; the proofs do not use the block structure).

9 Contrasts of composite operators

The operators of Table 1 enter the inequalities through composites: sums $A+B$ (items 2, 10, 12, 13), congruences $B^{1/2}AB^{1/2}$ (items 6, 7), means (item 6), Jordan products $\operatorname{Re} PQ = \frac{1}{2}(PQ+QP)$ (items 9, 21, 22), and, behind every characterisation, the commutator $AB - BA$, on which a functional vanishes if and only if it is a trace. This section describes how the contrast of a composite depends on the contrasts of its factors. Two reparametrisations of the contrast make the answers additive. For $A > 0$ with extreme eigenvalues $M \geq m > 0$ let

$$\eta(A) = \operatorname{artanh} \Delta(A) = \tfrac{1}{2} \log \frac{M}{m}, \qquad \psi(A) = \arcsin \Delta(A) \in [0, \tfrac{\pi}{2}], \qquad m(A) = \frac{M+m}{2}.$$

η is half the logarithm of the condition number. ψ is the *angle* of the operator in the sense of Krein and Gustafson [20, 21]: the largest angle between x and Ax, $\cos\psi(A) = \min_{x\neq 0}\langle Ax, x\rangle/(\|Ax\|\|x\|) = 2\sqrt{Mm}/(M+m)$, and $\sin\psi(A) = \min_{t>0}\|tA - 1\| = \Delta(A)$. For the density, $\kappa = \cosh\eta(W) = \sec\psi(W)$, so the envelope $\kappa - 1$ of the Kantorovich class (Theorem 3.1) is $\sec\psi(W) - 1$. The angle between vectors is $\angle(x, y) = \arccos(\operatorname{Re}\langle x, y\rangle/(\|x\|\|y\|))$, the angle of the real inner product $\operatorname{Re}\langle\cdot,\cdot\rangle$ on $\mathbb{C}^n = \mathbb{R}^{2n}$; it satisfies the triangle inequality.

Proposition 9.1 (Multiplicative composites: rapidities add)**.** *Let $A, B > 0$ in M_n and $\eta_A = \eta(A)$, $\eta_B = \eta(B)$.*

(a) $\Delta(A^p) = \tanh(|p|\eta_A)$ *for real $p \neq 0$,* $\Delta(A \otimes B) = \tanh(\eta_A + \eta_B)$*, and the compression of A to a subspace has contrast at most $\Delta(A)$.*

(b) $\tanh|\eta_A - \eta_B| \leq \Delta(A^{1/2}BA^{1/2}) \leq \tanh(\eta_A + \eta_B)$*; the same bounds hold for the contrast of $|AB|$, and for the spectrum of AB.*

(c) For the geometric mean $A\#B = A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2}$, $\tanh\frac{|\eta_A-\eta_B|}{2} \leq \Delta(A\#B) \leq \tanh\frac{\eta_A+\eta_B}{2}$.

(d) $\Delta(sA + tB) \leq \max(\Delta(A), \Delta(B))$ *for $s, t > 0$, and the same holds for the harmonic mean. If $\Delta(A), \Delta(B) > 0$, then $\Delta(A + tB) = 0$ for suitable $t > 0$ and a suitable relative position.*

All bounds are attained by commuting operators, with the extreme eigenvectors aligned (upper bounds) or interchanged (lower bounds), in every dimension $n \geq 2$.

Proof. (a) Spectral mapping, the eigenvalues of $A \otimes B$ are the products of eigenvalues, and Cauchy interlacing. (b) From $m_B A \leq A^{1/2}BA^{1/2} \leq M_B A$ and the monotonicity of the eigenvalues [3], $\lambda_k(A)m_B \leq \lambda_k(A^{1/2}BA^{1/2}) \leq \lambda_k(A)M_B$ for every k. Hence the largest eigenvalue lies in

$[M_A m_B, M_A M_B]$ and the smallest in $[m_A m_B, m_A M_B]$, so the condition number lies in $[h_A/h_B, h_A h_B]$; since $A^{1/2}BA^{1/2}$ is similar to $B^{1/2}AB^{1/2}$, it is also at least h_B/h_A. With $\Delta = \tanh(\frac{1}{2}\log h)$ this is (b). For $|AB|$ use the singular values, $\sigma_k(A)\sigma_{\min}(B) \le \sigma_k(AB) \le \sigma_k(A)\sigma_{\max}(B)$; AB is similar to $A^{1/2}BA^{1/2}$. (c) The mean is monotone and homogeneous, so $\sqrt{m_A m_B} \le A\#B \le \sqrt{M_A M_B}$, which gives the upper bound. From $B = (A\#B)A^{-1}(A\#B)$, whose spectrum is that of $A^{-1/2}(A\#B)^2A^{-1/2}$, (a) and (b) give $\eta_B \le \eta_A + 2\eta(A\#B)$, and symmetrically. (d) The mediant inequality $\frac{sM_A+tM_B}{sm_A+tm_B} \le \max(h_A, h_B)$ bounds the condition number of $sA+tB$; for the harmonic mean apply it to $A^{-1}+B^{-1}$. For the last claim take $B = c - A$ with $c > M_A$ in the eigenbasis of A and with the intermediate eigenvalues reflected; $\Delta(c-A) = (M_A - m_A)/(2c - M_A - m_A)$ takes every value in $(0,1)$. Attainment: for commuting A, B with aligned extreme eigenvectors all extreme eigenvalues multiply (add, for sums), and with interchanged ones the quotients do. □

The Jordan product. For $AB + BA$ the picture changes: it need not be positive, and angles, not rapidities, add. The reason is geometric. A positive operator A turns every vector by at most its angle: $\angle(x, Ax) \le \psi_A$, where $\sin\psi_A = \Delta(A)$. Hence ABx and BAx differ from x by at most $\psi_A + \psi_B$, and so does their sum $(AB+BA)x$, as long as $\psi_A + \psi_B < \frac{\pi}{2}$. Then $(AB+BA)x$ stays in the half-space of x, i.e. $\langle (AB+BA)x, x\rangle > 0$. If $\psi_A + \psi_B > \frac{\pi}{2}$, a suitable rotation of B against A turns some x by more than a right angle, and $AB+BA$ has a negative eigenvalue. In terms of contrasts: *$AB+BA$ is positive for every relative position if and only if the point $(\Delta(A), \Delta(B))$ lies in the unit disc* (Figure 3). This criterion is Gustafson's [20]; we add that it is sharp in every dimension and that the same angles bound the contrast of $AB+BA$.

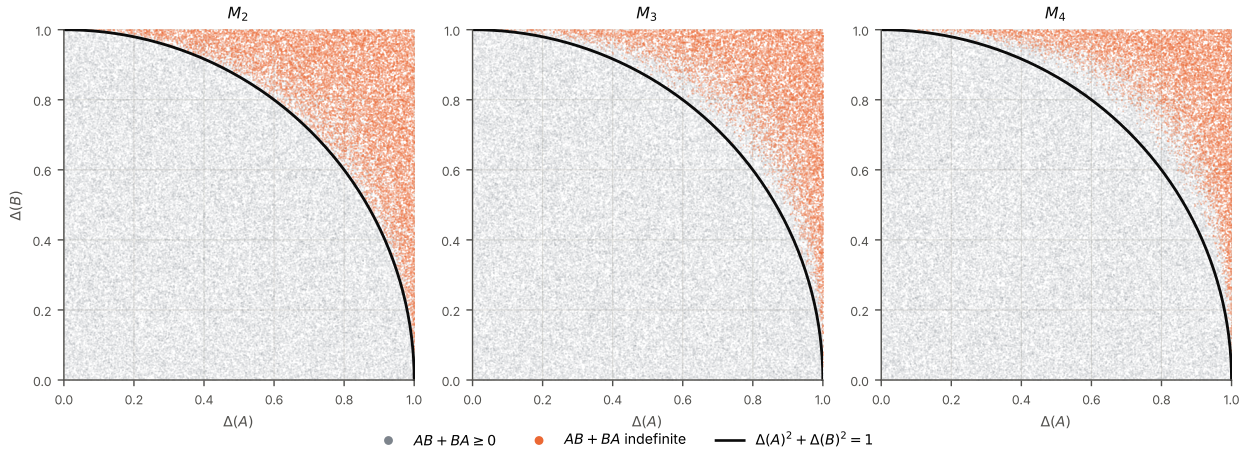


Figure 3: Is $AB+BA$ positive? 100000 random pairs $A, B > 0$ per dimension, with contrasts uniform in $[0,1]^2$ and Haar random relative position: positive (grey) or indefinite (orange). No indefinite product occurs inside the quarter circle $\Delta(A)^2 + \Delta(B)^2 = 1$ (Theorem 9.2(a)); outside it occurs for some positions, more often the further out.

Theorem 9.2 (Jordan product). *Let $A, B > 0$ in M_n, $n \ge 2$, with contrasts δ_A, δ_B, angles $\psi_A = \arcsin\delta_A$, $\psi_B = \arcsin\delta_B$ and rapidities $\eta_A = \operatorname{artanh}\delta_A$, $\eta_B = \operatorname{artanh}\delta_B$.*

(a) $AB+BA \ge 0$ for every relative position of A and B if and only if $\delta_A^2 + \delta_B^2 \le 1$. If $\delta_A^2 + \delta_B^2 > 1$, $AB+BA$ is indefinite for some relative position, in every dimension.

(b) If $\delta_A^2 + \delta_B^2 < 1$, then $\tanh|\eta_A - \eta_B| \le \Delta(AB+BA) \le \sin(\psi_A + \psi_B)$, and the lower bound is attained by commuting A, B whose extreme eigenvectors are interchanged.

Proof. (a) and the upper bound in (b). We use the real angle $\angle(x,y)$ defined above. By the formula for $\cos\psi$, $\angle(x,Bx)\le\psi_B$ and $\angle(Bx,ABx)\le\psi_A$, so $\angle(x,ABx)\le\psi_A+\psi_B$ by the triangle inequality; likewise $\angle(x,BAx)\le\psi_A+\psi_B$. If $\psi_A+\psi_B<\frac{\pi}{2}$, the cone $\{y:\angle(x,y)\le\psi_A+\psi_B\}$ is convex and contains $(AB+BA)x$, so $\langle(AB+BA)x,x\rangle>0$; moreover the largest angle of $AB+BA$ is at most $\psi_A+\psi_B$, and its sine is $\Delta(AB+BA)$. The case $\psi_A+\psi_B=\frac{\pi}{2}$ follows by continuity.

Converse in (a). On M_2 take $A=\operatorname{diag}(1,a)$ and B with eigenvalues $1,b$ and eigenvector $(\cos\theta,\sin\theta)$ for 1, where $a=\frac{1-\delta_A}{1+\delta_A}$, $b=\frac{1-\delta_B}{1+\delta_B}$. With $c=\cos^2\theta$ and $p=(1-a)(1-b)$ a direct computation gives

$$\tfrac{1}{2}\operatorname{Tr}(AB+BA)=a+b+pc,\qquad \det(AB+BA)=4ab-p^2c(1-c).$$

At $\theta=\pi/4$ the determinant is negative if and only if $16ab<p^2$; with $\sqrt{a}=\sqrt{1-\delta_A^2}/(1+\delta_A)$ and $1-a=2\delta_A/(1+\delta_A)$ this reads $(1-\delta_A^2)(1-\delta_B^2)<\delta_A^2\delta_B^2$, i.e. $\delta_A^2+\delta_B^2>1$. On M_n add the direct summands $a\,1_{n-2}$ and $b\,1_{n-2}$.

Lower bound in (b). If x,y are unit eigenvectors of A for its extreme eigenvalues M_A, m_A, then $\lambda_{\max}(AB+BA)\ge 2\operatorname{Re}\langle Bx,Ax\rangle=2M_A\langle Bx,x\rangle\ge 2M_Am_B$ and likewise $\lambda_{\min}(AB+BA)\le 2m_AM_B$. So the condition number of $AB+BA$ is at least h_A/h_B, and by symmetry at least h_B/h_A. □

How large can $\Delta(AB+BA)$ be? The upper bound $\sin(\psi_A+\psi_B)$ is reached only at the boundary of the disc. The true maximum depends on the dimension, because in M_2 both extreme eigenvalues of $AB+BA$ come from the same pair of 2×2 matrices, while in M_3 they can come from different blocks.

Proposition 9.3 (Largest contrast of the Jordan product)**.** *In the notation of the proof above, let* $\delta_A^2+\delta_B^2<1$.

(a) On M_2, $\max\Delta(AB+BA)=\max\limits_{0\le c\le 1}\Big(1-\dfrac{4ab-p^2c(1-c)}{(a+b+pc)^2}\Big)^{1/2}$.

(b) On M_n, $n\ge 3$, $\max\Delta(AB+BA)\ge(2-\mu)/(2+\mu)$, *where* $\mu=\min_{0\le c\le 1}\big(a+b+pc-\sqrt{(a+b+pc)^2-4ab+p^2c(1-c)}\,\big)$ *is the smallest eigenvalue of* $AB+BA$ *over the* 2×2 *pairs of (a).*

Numerically, both values equal the commuting value $\tanh(\eta_A+\eta_B)$ *for small contrasts and exceed it for large ones.*

Proof. (a) On M_2 every pair is unitarily equivalent to the pair of the proof of Theorem 9.2 for some θ, and $\Delta^2=1-4\det/\operatorname{Tr}^2$. (b) Take $A=1\oplus A_2$ and $B=1\oplus B_2$ (padded with ones) with the 2×2 pair A_2,B_2 of smallest eigenvalue μ. Then $AB+BA=2\oplus(A_2B_2+B_2A_2)$ has the largest eigenvalue 2, since $\|A_2B_2+B_2A_2\|\le 2$, and the smallest eigenvalue μ. □

For $\delta_A=\delta_B=0.6$: the commuting value is 0.8824, the maximum on M_2 is 0.8839, the value of (b) for $n\ge 3$ is 0.8963, and $\sin(\psi_A+\psi_B)=0.9600$. A numerical optimisation over all relative positions and intermediate eigenvalues (`scripts/contrast_operations.py`, 24 starts per point) finds on M_3 and on M_4 exactly the value of (b) (largest difference 3×10^{-7} on a grid of δ_A with $\delta_B=0.6$; Figure 7). Equality for every $n\ge 3$ would follow if the smallest eigenvalue of a Jordan product with spectra in $[a,1]$ and $[b,1]$ were always minimised on a two-dimensional subspace, the setting of Strang's eigenvalue bounds [33]; we do not use this.

The theorem explains a threshold. The analogue of item 21 for positive operators, $\varphi(\operatorname{Re}AB)\ge 0$, holds for every positive functional as long as $\Delta(A)^2+\Delta(B)^2\le 1$, because then $\operatorname{Re}AB\ge 0$; for operators of the same contrast δ' it can fail only for $\delta'>1/\sqrt{2}$. For projections, $\psi=\frac{\pi}{2}$, and $\operatorname{Re}PQ$ is

indefinite in general; this is why item 21 is violated by every non-tracial functional, with the envelope $\frac{1}{2}\tan^2(\psi/2)$ of Theorem A.

Proposition 9.4 (Commutators: three contrasts multiply)**.** *Let $A, B > 0$ in M_n.*

(a) $\|AB-BA\| \le 2\Delta(A)\Delta(B)\,m(A)m(B)$*, with equality for suitable A, B of any prescribed contrasts in every M_n, $n \ge 2$. In particular* $\|AB-BA\| \le \frac{2\Delta(A)\Delta(B)}{(1+\Delta(A))(1+\Delta(B))}\|A\|\|B\| \le \frac{1}{2}\|A\|\|B\|$.

(b) For a state φ_W ($\operatorname{Tr} W = 1$) with $\Delta(W) = \delta$, $h = \frac{1+\delta}{1-\delta}$,

$$|\varphi_W(AB-BA)| \le 2\,c_n(\delta)\,\Delta(A)\Delta(B)\,m(A)m(B), \qquad c_n(\delta) = \begin{cases} \delta, & n \text{ even}, \\ \dfrac{k(h-1)}{k(h+1)+1}, & n = 2k+1, \end{cases}$$

and the bound is attained for every δ and all prescribed contrasts of A and B.

Proof. (a) Write $A = m(A)(1+E)$, $B = m(B)(1+F)$ with self-adjoint E, F, $\|E\| = \Delta(A)$, $\|F\| = \Delta(B)$. Then $AB-BA = m(A)m(B)(EF-FE)$ and $\|EF-FE\| \le 2\|E\|\|F\|$. Equality holds on M_2 for $E = \Delta(A)\sigma_z$, $F = \Delta(B)\sigma_x$, since $\sigma_z\sigma_x - \sigma_x\sigma_z = 2i\sigma_y$; in M_n add a direct summand $m(A)\oplus m(B)$. The second form uses $\|A\| = m(A)(1+\Delta(A))$. (b) $C = i(AB-BA)$ is self-adjoint with $\operatorname{Tr} C = 0$ and $\|C\| \le K$, the bound of (a). By von Neumann's trace inequality [3], $|\operatorname{Tr} WC|$ is at most the maximum of $\sum_i w_i c_i$ over decreasingly ordered w_i (the eigenvalues of W) and $c_i \in [-K, K]$ with $\sum_i c_i = 0$, which is $K\sum_{i\le k}(w_i - w_{n+1-i})$, $k = \lfloor n/2 \rfloor$. Let $a = w_1$, $b = w_n = a/h$ and $S = \sum_{i\le k} w_i \le ka$. Then $\sum_{i\le k}(w_i - w_{n+1-i}) \le S - kb$ and $1 = \operatorname{Tr} W \ge S + (n-k)b$. Maximising $S - kb$ under $S \le khb$ and $S \le 1-(n-k)b$ gives the value at $khb = 1-(n-k)b$, namely $k(h-1)/(kh+n-k)$, which is $c_n(\delta)$ for even and odd n. For equality take k copies of the 2×2 pair of (a), W diagonal in the eigenbasis of the k blocks of C with the eigenvalues a on the positive and b on the negative eigenvectors of C, and, for odd n, the eigenvalue b of W and scalars A, B on the remaining one-dimensional block. □

So the most basic characterisation of the trace, $\varphi(AB) = \varphi(BA)$, fails at the order of the product of three contrasts, $2\Delta(W)\Delta(A)\Delta(B)$ in even dimension: the thesis (I) in its simplest form. The odd dimensions are slightly weaker, as for Gardner's inequality (Theorem 4.1), and $c_{2k+1}(\delta) \to \delta$ as $k \to \infty$.

Figures 4–8 compare random samples (100000 per figure and dimension, Haar random relative positions, the intermediate eigenvalues uniform between the extreme ones) with these bounds on M_2, M_3 and M_4. No sample exceeds a bound (the largest value of sample minus bound is -5×10^{-8}). Among 100000 further random pairs per dimension with contrasts uniform in $[0,1]^2$, the number of indefinite Jordan products with $\delta_A^2 + \delta_B^2 \le 1$ is 0. As in Section 10.2, random positions rarely come close to the extremes in higher dimension: for the commutator test on M_4 the samples stay far below the bound, which needs two aligned 2×2 blocks.

10 Numerical verification and the remaining items

10.1 Method

All envelopes were computed by constrained optimisation with our PyTorch library `torch_vn_algebra` [26]; the program is `scripts/trace_characterizations.py` of the repository [26]. The density W is diagonal with a spectrum parametrised so that $\Delta(W) = \delta$ holds exactly and the other eigenvalues are free. Positive variables have a free contrast in $[0,1]$, so rank-deficient operators are included; there are self-adjoint and unitary variables (polar factors, and projections UDU^*, with

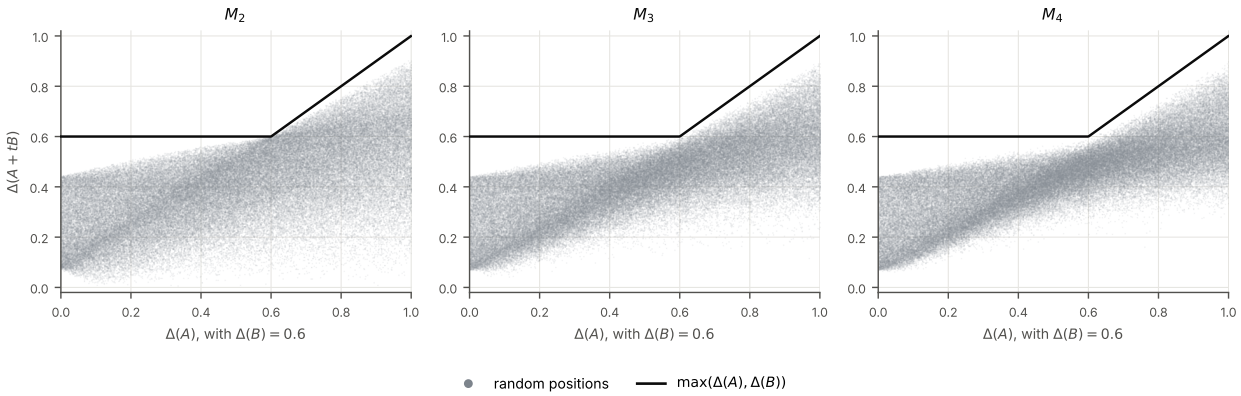


Figure 4: Contrast of $A + tB$ (t log-uniform) against $\Delta(A)$, with $\Delta(B) = 0.6$: it never exceeds $\max(\Delta(A), \Delta(B))$ (Proposition 9.1(d)).

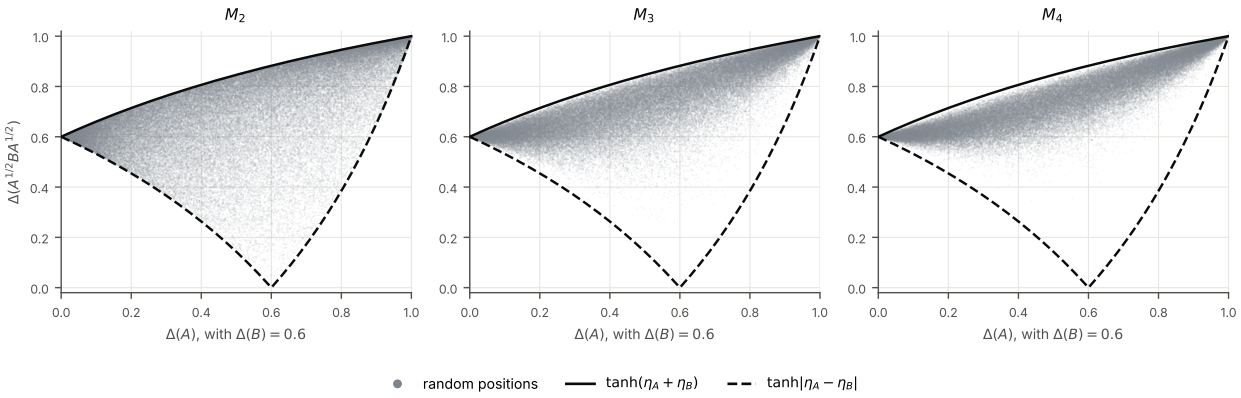


Figure 5: Contrast of $A^{1/2}BA^{1/2}$ against $\Delta(A)$, with $\Delta(B) = 0.6$, and the bounds $\tanh|\eta_A \pm \eta_B|$ of Proposition 9.1(b).

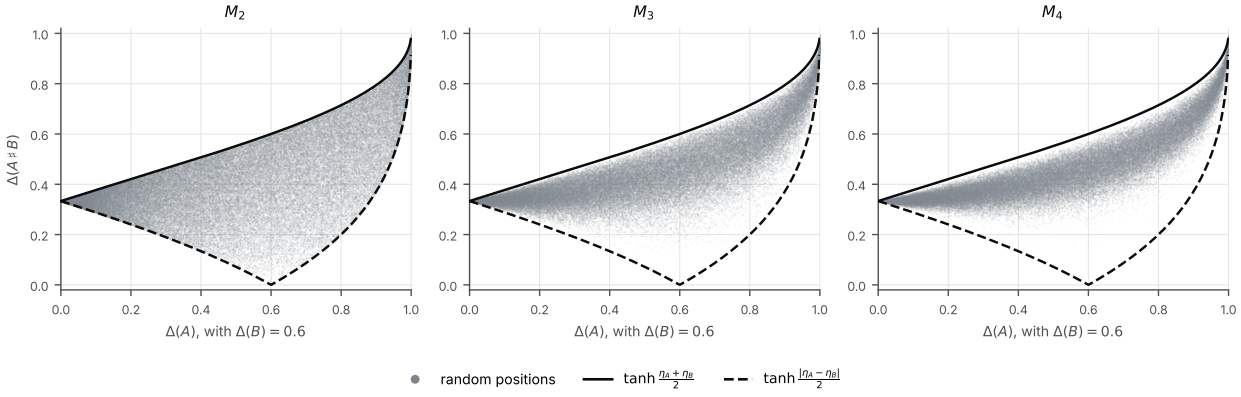


Figure 6: Contrast of the geometric mean $A\#B$ against $\Delta(A)$, with $\Delta(B) = 0.6$, and the bounds $\tanh\frac{1}{2}|\eta_A \pm \eta_B|$ of Proposition 9.1(c).

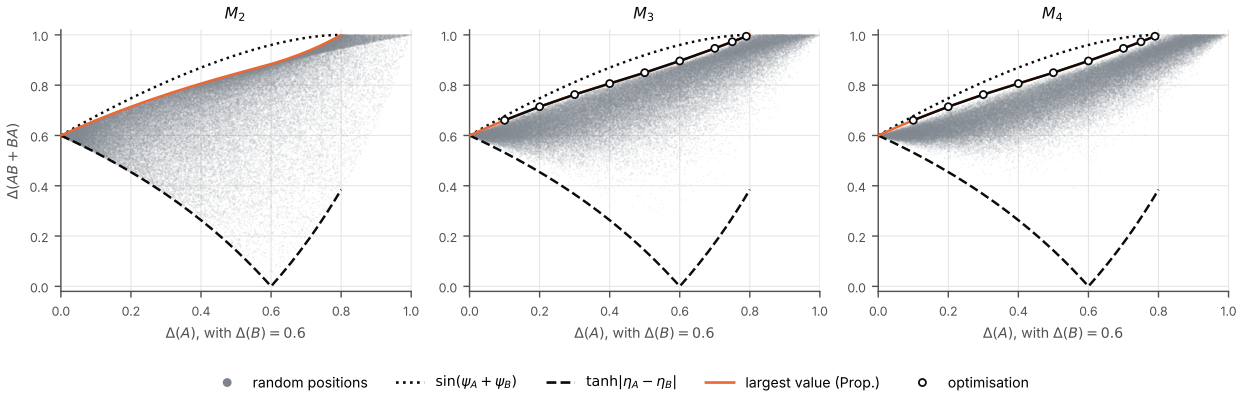


Figure 7: Contrast of the Jordan product $AB + BA$ against $\Delta(A) < 0.8$, with $\Delta(B) = 0.6$ (for $\Delta(A) \leq 0.8$ it is positive, Theorem 9.2(a)): the bounds of Theorem 9.2(b), the largest value of Proposition 9.3 (orange; on M_3 and M_4 the value $(2-\mu)/(2+\mu)$), and the largest value found by optimisation (markers).

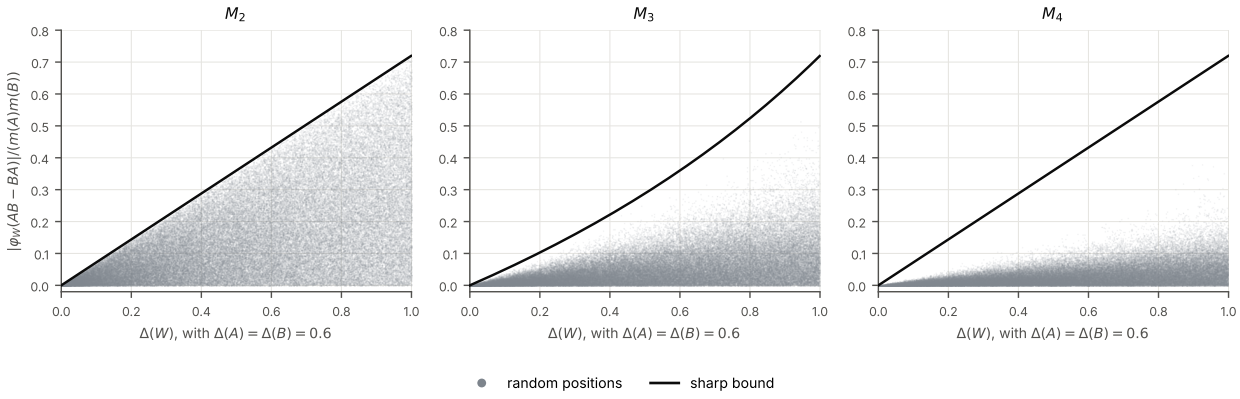


Figure 8: The commutator test $|\varphi_W(AB - BA)|/(m(A)m(B))$ for states φ_W against $\Delta(W)$, with $\Delta(A) = \Delta(B) = 0.6$, and the sharp bound $2c_n(\delta) \cdot 0.36$ of Proposition 9.4(b).

every rank pattern spread over the starts), and positive scalars. Half of the starts begin at structured points (two-level spectra, rotations in pairs of eigenvectors of W). The lattice operations are the spectral projections of $P+Q$ for the eigenvalue 2 ($P \wedge Q$) and for the range ($P \vee Q$), computed by functional calculus with an indicator function, whose Daleckii–Krein gradient is the derivative of the spectral projection. In 19 and 23 the suprema are approached as P and Q merge, where $P \wedge Q$ and $P \vee Q$ jump; there the search maximises $(L-R)/N - 10^{-12}/\|P-Q\|_{\mathrm{F}}^2$, which keeps P and Q about 10^{-3} apart, while the reported values are those of $(L-R)/N$. A grid of 14 contrasts times 64 starts forms one batch per item and dimension (800 gradient steps, $n = 2, 3, 4$).

Certification. The best final point of every contrast is recomputed in mpmath with 40 digits, after the unitaries are re-orthonormalised and the contrast is recomputed from the stored eigenvalues; the lattice operations are recomputed from the exact eigenvalues. Every positive value reported below is thus a certified violation at an exactly admissible point. The double-precision and mpmath values agree to 5×10^{-6} (relative), and at $\delta = 0$ no value exceeds 1×10^{-16}. Independently, plain NumPy/SciPy searches (BFGS from 40 random starts on M_2 at $\delta = 0.3, 0.5, 0.9$) give values whose ratio to the batched search lies in $[0.994, 1.097]$; ratios above 1 come from item 13, where BFGS stops earlier. The test suite compares every objective with a plain NumPy implementation and with mpmath, checks that random points stay below the proved envelopes and that the stated extremisers attain them. The whole computation takes 475 s on 4 CPU processes.

10.2 Random sampling

Before searching for the extremes we sample. A third program (`scripts/trace_montecarlo.py`, plain NumPy, independent of the library) draws, for each of the 25 items and for $n = 2, 3, 4$, 200000 random functionals on M_n and random operators: W with a uniformly distributed contrast $\Delta(W) \in [0, 1)$ and the other eigenvalues uniform in between, positive operators $U\mathrm{diag}(\lambda)U^*$ with Haar-random U, a uniformly distributed contrast and a random scale, arbitrary operators $V|X|$ with a Haar-random V, self-adjoint ones with random signs, and Haar-random projections. This is the method of [1, Sect. 5], extended to all items. A second sample replaces every positive operator (every modulus) by a scaled projection of random rank (*projection sample*). Appendix A has one figure per item with one panel per dimension: every sample, $(L-R)/N$ against $\Delta(W)$, together with the certified envelope $V(\delta)$. For the 14 items with operator variables a second row shows a third sample, with $\Delta(W) = 0.9$ and all operator variables of one contrast δ', against δ', together with the certified $V(0.9, \delta')$ of Section 10.4. Table 2 gives the fraction of violating samples.

The samples show four things.

(a) *No sample exceeds the envelope*: the largest ratio of a sampled violation to the certified envelope at the upper edge of its bin is 1.003 over all items and dimensions (the excess over 1 is the linear interpolation between grid values of the envelope). This is an independent check of the computed envelopes.

(b) *Random operators rarely come close to the envelope*, and for monotonicity, subadditivity, the Powers–Størmer inequality and the weighted monotonicity essentially never violate it: less than 0.1% of the samples (or none) violate. A Monte Carlo study alone would conclude that these inequalities hold for every φ_W.

(c) *Sampling gets worse with the dimension*: for Gardner's inequality the fraction of violating random samples drops from 4.4% on M_2 to below 0.1% on M_3 and on M_4, and the best sample stays far below the envelope (Figure 9). The extremal configurations, rank-one operators on a two-dimensional subspace aligned with the extreme eigenvectors of W, have a Haar measure that decreases rapidly with n.

item	inequality	M_2 random	M_2 proj.	M_3 random	M_3 proj.	M_4 random	M_4 proj.
1	Gardner	4.4	19.0	< 0.1	2.9	< 0.1	0.6
2	triangle	1.1	6.3	< 0.1	6.2	< 0.1	4.5
3	$\varphi(A{+}A^*)$	0.6	3.1	< 0.1	0.4	0	< 0.1
4	$\lvert\varphi(A^2)\rvert$	6.6	6.8	0.2	0.4	< 0.1	< 0.1
5	metric	0	–	< 0.1	–	< 0.1	–
6	geom. mean	< 0.1	–	0	–	0	–
7	ALT	19.5	0	14.3	0	11.0	0
8	Young (ext.)	7.6	8.4	3.0	2.1	1.4	0.6
9	Bikchentaev-Abed	50.2	–	50.1	–	49.8	–
10	monotonicity	< 0.1	0.7	0	< 0.1	0	< 0.1
11	AM-GM	20.5	–	13.2	–	10.0	–
12	subadditivity	< 0.1	21.3	0	9.1	0	5.1
13	Powers-Stormer	< 0.1	9.0	0	3.0	0	1.4
14	Hoelder	1.3	–	< 0.1	–	0	–
15	Golden-Thompson	25.6	–	16.8	–	11.5	–
16	Peierls-Bogoliubov	2.4	1.8	0.8	0.4	0.4	0.1
17	$\varphi(PQP)$	12.0	–	4.0	–	1.9	–
18	$\varphi(\lvert P-Q\rvert)$	13.0	–	4.3	–	2.1	–
19	$P \wedge Q$	49.9	–	37.8	–	33.3	–
20	e^{QPQ}	8.0	–	2.2	–	1.0	–
21	$\mathrm{Re}PQ \geq 0$	5.6	–	1.4	–	0.6	–
22	$\mathrm{Re}PQ \leq \lvert PQ\rvert$	14.6	–	7.7	–	4.3	–
23	$P \vee Q$	49.8	–	37.7	–	33.5	–
24	power mean	7.2	50.0	0.2	25.1	< 0.1	16.8
25	weighted mon.	0	0	0	0	0	0

Table 2: Random sampling, 200000 samples per item, class and dimension: the percentage of samples that violate the inequality, for random operators and for scaled projections. "–": not applicable (the variables are projections, or an invertible operator is needed).

(d) *Projections detect the failure.* Sampling scaled projections instead of general positive operators raises the fraction of violating samples in every dimension, for Gardner's inequality from 4.4% to 19% on M_2 and from below 0.1% to 0.6% on M_4, for subadditivity from below 0.1% to 21%, 9% and 5% on M_2, M_3, M_4, for the power-mean inequality from 7% to 50% on M_2, and brings the best sample close to the envelope (orange points in Figures 9, 20 and 32). The exceptions are those of Section 10.3: for items 4 and 7 the extremisers are two-level operators that are not projections (for item 7 the violation degenerates exactly at projections), and items 6 and 11 need invertible operators.

10.3 Do projections suffice?

For the 14 items whose variables are not projections, a second program (`scripts/trace_contrast.py`) repeats the search with the positive variables (the moduli, for the others) restricted in two ways: to two-level operators $c(\ell + (1-\ell)P)$ with free $\ell \in [0,1]$, $c > 0$ and a projection P of any rank (for self-adjoint variables $\alpha + \beta P$ with free real α, β), and to scaled projections cP. Everything else is as before (64 starts per grid point, 800 gradient steps, best final point recomputed in mpmath with 40 digits); the whole computation of this and the next subsection takes 760 s on four CPU processes.

Table 3 shows four groups.

	M_2						M_3					
	$\delta = 0.5$			$\delta = 0.9$			$\delta = 0.5$			$\delta = 0.9$		
item	all	two-level	proj.	all	two-level	proj.	all	two-level	proj.	all	two-level	proj.
1	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942
2	0.1547	0.1547	0.1547	1.2942	1.2941	1.2942	0.1547	0.1547	0.1547	1.2942	1.2941	1.2942
3	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942
4	0.1547	0.1547	0.0774	1.2942	1.2942	0.6471	0.1547	0.1328	0.0774	1.2942	1.2612	0.7393
6	0.1427	0	0.1501	1.2139	1.1707	1.2924	0.1359	0	0.1491	1.2125	1.1405	1.2922
7	0.2656	0.2730	0	3.4387	3.5241	0	0.2733	0.2703	0	3.4965	3.4358	0
8	2.0000	2.0000	2.0000	18.00	18.00	18.00	2.0000	2.0000	2.0000	18.00	18.00	18.00
10	0.0052	0.0052	0.0052	0.1825	0.1825	0.1825	0.0052	0.0052	0.0052	0.1825	0.1825	0.1825
11	0.1340	0.1340	–	0.5641	0.5641	–	0.1347	0.1347	–	0.6188	0.6188	–
12	0.1338	0.1338	0.1338	0.9678	0.9679	0.9680	0.1338	0.1338	0.1338	0.9678	0.9675	0.9680
13	0.0911	0.0329	0.0911	0.2197	0.2171	0.2197	0.0911	0.0911	0.0911	0.3048	0.2057	0.2197
16	0.0560	0.0561	0.0560	0.8885	0.8885	0.8873	0.0561	0.0561	0.0560	0.8894	0.8887	0.8864
24	0.4142	0.4142	0.4142	2.1623	2.1623	2.1623	0.5275	0.5275	0.5275	2.6056	2.6056	2.6056
25	$1.6 \cdot 10^{-6}$	$1.4 \cdot 10^{-6}$	0	0.0017	0.0017	0	$5.3 \cdot 10^{-7}$	$2.2 \cdot 10^{-8}$	0	0.0017	0.0017	0

Table 3: Certified largest violations over all operators, over two-level operators and over scaled projections, for the items whose variables are not projections. In M_2 every operator is two-level, so there the middle column only checks the restricted parametrisation. "–": item 11 needs an invertible A.

(a) *Projections attain the envelope*: items 1, 2, 3, 8, 10, 12, 16 and 24 in M_2 and M_3 (item 16 to within 0.3%), and item 13 in M_2. Together with the 11 items whose variables are projections by definition, this covers 20 of the 25 items.

(b) *Projections approach it*: in items 6 and 7 the supremum is approached as an operator becomes a projection, where both sides of the inequality degenerate. Restricted to projections (item 6) or to two-level operators (item 7), the search comes closer to the proved lower bound $\kappa - 1$, resp. $\kappa^2 - 1$, than without restriction: the restriction helps the search.

(c) *Two-level operators are needed*: item 4 (the extremiser $A = \sqrt{a}\, e_1 e_2^* + \sqrt{b}\, e_2 e_1^*$ has $|A|$ with the eigenvalues $\sqrt{a}, \sqrt{b}$), item 11 ($A = W^{-1}$) and item 25 (A with the eigenvalue ratio $\approx \delta^2/8$, Proposition 7.2; rank-one A gives no violation). In M_3 the extremiser of item 4 lives on a two-dimensional subspace and vanishes on its complement, so it has three eigenvalues $\sqrt{a}, \sqrt{b}, 0$; the two-level class without 0 misses it.

(d) *Exceptions*: in M_3 at $\delta = 0.9$ the generalised Powers–Størmer inequality (item 13) is violated by 0.3048 against 0.2197 for projections and for M_2, and the spectral geometric mean (item 6) exceeds $\kappa - 1$ for $n \geq 3$ (Section 10, item (iii) below). Here operators that are not two-level do more.

10.4 Both contrasts

The bottom rows of Figures 9–33 (Appendix A) show $V(0.9, \delta')$ for the same 14 items, with random samples. At $\delta' = 0$ no item is violated (Proposition 2.4). Beyond that, the items fall into four classes.

(a) *Smooth growth in both contrasts*: items 1–3 follow Theorem 4.1 to 1×10^{-9} (relative, $n = 2, 3$, all grid points); item 8 follows Proposition 5.3 on M_2.

(b) *A threshold in the contrast of the operators*: items 6, 10, 12, 13, 24 and 25 show no violation until δ' is close to 1; at $\delta = 0.9$ the first grid value with a violation above 10^{-4} is: 6 ($n = 2$): 0.9; 6 ($n = 3$): 0.8; 10 ($n = 2$): 0.8; 10 ($n = 3$): 0.8; 12 ($n = 2$): 0.99; 12 ($n = 3$): 0.99; 13 ($n = 2$): 0.9; 13 ($n = 3$): 0.9; 24 ($n = 2$): 0.8; 24 ($n = 3$): 0.9; 25 ($n = 2$): –; 25 ($n = 3$):

0.99. These characterisations are blind to operators of moderate contrast: their violation needs nearly singular operators.

(c) *A peak*: item 11 is violated most strongly when the contrast of A is close to that of W (its extremiser is W^{-1}), and in M_2 items 4 and 7 have an interior maximum in δ' (for item 4 at the contrast of $|A|$ of the extremiser in (c) above); in M_3 the additional dimension removes the maxima of 4 and 7.

(d) *No dependence on the contrast of the operators*: item 16 is invariant under $A \mapsto A + c$, $B \mapsto B + c$, which makes the contrasts of A and B arbitrarily small without changing the quotient; accordingly $V(\delta, \delta')$ equals the envelope with free operators for every $\delta' > 0$ and jumps to 0 at $\delta' = 0$.

On M_2 every spectrum is two-level. On M_3 the two-level operators of contrast δ' reproduce the values of general spectra with the same contrast except in items 4, 6 and 7 near $\delta' = 1$: there the extremisers need a third eigenvalue (a zero on the complement of a two-dimensional subspace, or a nearly singular second operator).

10.5 Envelopes with the operators free

items	$V^\star(\delta)$	as $\delta \to 0$	$V^\star(0.5)$	search $V(0.5)$	status
8	$h - 1 = 2\delta/(1-\delta)$	2δ	2.0000	2.0000	sharp
9	$s/(1+s)$, $s = \sin\frac{\psi}{2}$	$\delta/2$	0.2056	0.2056	sharp for $n = 2$
19	δ	δ	0.5000	0.5000	sharp, approached
23	$\delta/(1-\delta)$	δ	1.0000	1.0000	sharp, approached
24	$(1-\delta)^{-1/2} - 1$	$\delta/2$	0.4142	0.4142	sharp for $n = 2$; grows with n
1–5, 18, 22	$\kappa - 1$	$\delta^2/2$	0.1547	0.1547	sharp
17	$\kappa^2 - 1$	δ^2	0.3333	0.3333	sharp
11	$1 - \sqrt{1-\delta^2}$	$\delta^2/2$	0.1340	0.1340	sharp for $n = 2$; grows with n
20	max. of one variable	$\frac{(e-1)^2}{e(e+1)}\delta^2$	0.0802	0.0802	sharp
21	$\frac{1}{2}\tan^2(\psi/2)$	$\delta^2/8$	0.0359	0.0359	sharp
6	$\geq \kappa - 1$	$\geq \delta^2/2$	0.1547	0.1430	lower bound
7	$\geq \kappa^2 - 1$	$\geq \delta^2$	0.3333	0.2653	lower bound
12–16	–	$c\,\delta^2$, $c \in [\frac{1}{36}, \frac{1}{2}]$	–	0.0086–0.1338	search
10	$\geq m(\delta)$	$\geq \delta^4/16$	0.0052	0.0052	lower bound
25	$\geq m_{25}(\delta)$	$\geq \delta^8/4096$	1.7×10^{-6}	1.5×10^{-6}	lower bound

Table 4: Classes of behaviour, ordered by the order of vanishing of the envelope at the trace. $V^\star$: the sharp value or, where marked $\geq$, a proved lower bound; "search": the certified search value on M_2 (a range over the items of a row). Items 11 and 24 are given for $n = 2$.

Tables 4 and 5 and the top rows of Figures 9–33 summarise the computation.

(i) *Agreement with the theorems.* Where the envelope is known exactly and attained (items 1–5, 8, 11, and 9 for $n = 2$), the search reproduces it to 1×10^{-7} for $\delta \geq 0.05$ and $n = 2, 3, 4$; for items 17, 18, 20–22 and 24 to 3×10^{-4}, and for the approached suprema of 19 and 23 to 2×10^{-6}. Apart from item 6 with $n \geq 3$, no value exceeds the lower bounds of Section 7 (largest relative excess 1×10^{-16}). The values of 17, 18 and 21–23 on M_3 and M_4 agree with the M_2 values to 4×10^{-7}, as Theorem 6.1 predicts, and the search on M_3 finds the odd-n value 0.0694 of item 20 at $\delta = 0.5$ (M_2: 0.0802).

(ii) *Items without a closed form.* All five items 12–16 fail quadratically, with $V \approx 0.5\,\delta^2$ (12, 13), $\delta^2/36$ (14, equal to the bound of Proposition 7.1(d) to leading order), $0.064\,\delta^2$ (15) and $0.09\,\delta^2$

	inequality $L \le R$	ref.	$V(0.1)$	$V(0.5)$	$V(0.9)$	$V^\star(0.5)$	order	slope	n-dep.
1	$\lvert\varphi(X)\rvert \le \varphi(\lvert X\rvert)$	[18]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	4×10^{-10}
2	$\varphi(\lvert X+Y\rvert) \le \varphi(\lvert X\rvert)+\varphi(\lvert Y\rvert)$	[18, 32]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-9}
3	$\varphi(A+A^*) \le 2\varphi(\lvert A^*\rvert)$	[10]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	5×10^{-10}
4	$\lvert\varphi(A^2)\rvert \le \varphi(A^*A)$	[29, 14]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
5	$\varphi(\lvert P-R\rvert) \le \varphi(\lvert P-Q\rvert)+\varphi(\lvert Q-R\rvert)$	[11]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
6	$\varphi(A\natural B) \le \sqrt{\varphi(A)\varphi(B)}$	[15]	n.r.	0.1430	1.1979	0.1547^b	δ^2	–	$n\ge3$
7	$\varphi((B^{1/2}AB^{1/2})^2) \le \varphi(BA^2B)$	[9]	0.0059	0.2653	3.4322	0.3333^b	δ^2	2.31	9×10^{-2}
8	$\varphi(\lvert AB^*\rvert) \le \frac12\varphi(\lvert A\rvert^2)+\frac12\varphi(\lvert B\rvert^2)$	[4]	0.2222	2.0000	18.00	2.0000	δ^1	1.17	9×10^{-12}
9	$\varphi(PQ+QP) \le 2\varphi(QPQ)$	[10]	0.0477	0.2056	0.3469	0.2056^a	δ^1	0.94	3×10^{-10}
10	$0\le A\le B \Rightarrow \varphi(A^2)\le\varphi(B^2)$	[5, 30, 24]	6.3×10^{-6}	0.0052	0.1825	0.0052^b	δ^4	4.04	3×10^{-8}
11	$(\det A)^{1/n} \le \frac1n\varphi(A)$, $\varphi(1)=n$	[12]	0.0050	0.1340	0.5641	0.1340	δ^2	2.01	yes
12	$\varphi((A+B)^{1/2}) \le \varphi(A^{1/2})+\varphi(B^{1/2})$	[34]	0.0050	0.1338	0.9678	–	–	2.00	5×10^{-4}
13	$\varphi(A+B-\lvert A-B\rvert) \le 2\varphi(A^{1/4}B^{1/2}A^{1/4})$	[25]	0.0048	0.0911	0.2197	–	–	1.93	4×10^{-1}
14	$\lvert\varphi(PQ)\rvert \le \varphi(P)^{1/3}\varphi(Q)^{2/3}$	[8, 10]	2.8×10^{-4}	0.0086	0.0711	–	–	2.03	4×10^{-3}
15	$\varphi(e^{P+Q}) \le \varphi(e^{P/2}e^Qe^{P/2})$	[8]	6.4×10^{-4}	0.0142	0.0420	–	–	1.98	2×10^{-1}
16	$\varphi(e^A)e^{\varphi(e^{A/2}Be^{A/2})/\varphi(e^A)} \le \varphi(e^{A+B})$	[9]	9.7×10^{-4}	0.0561	0.8890	–	–	2.35	2×10^{-3}
17	$\varphi(PQP) \le \varphi(Q)$	[7]	0.0101	0.3333	4.2632	0.3333	δ^2	2.04	7×10^{-9}
18	$\varphi(\lvert P-Q\rvert) \le \varphi(P+Q)$	[7]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
19	$\varphi(P+Q) \le \varphi(1+P\wedge Q)$	[7]	0.1000	0.5000	0.9000	0.5000^c	δ^1	1.00	odd n
20	$\varphi(e^{QPQ}) \le \varphi(e^P)$	[8]	0.0029	0.0802	0.3115	0.0802	δ^2	2.02	odd n
21	$\varphi(\mathrm{Re}\,PQ) \ge 0$	[8]	0.0013	0.0359	0.1964	0.0359	δ^2	2.02	8×10^{-9}
22	$\varphi(\mathrm{Re}\,PQ) \le \varphi(\lvert PQ\rvert)$	[8]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	5×10^{-9}
23	$\varphi(P\vee Q) \le \varphi(P)+\varphi(Q)$	[29]	0.1111	1.0000	9.0000	1.0000^c	δ^1	1.17	4×10^{-7}
24	$\varphi(A^2)^{1/2} \le \varphi(A)$, $\varphi(1)=n$	[12]	0.0541	0.4142	2.1623	0.4142	δ^1	1.13	yes
25	$0\le A\le B \Rightarrow \varphi(A^4)\le\varphi(AB^2A)$	[24]	n.r.	1.5×10^{-6}	0.0017	1.7×10^{-6b}	δ^8	–	3×10^{-6}

Table 5: Certified search values on M_2 at $\delta = 0.1, 0.5, 0.9$ and the proved values $V^\star$; ^{a}: sharp for $n=2$ only, ^{b}: a lower bound, ^{c}: approached but not attained. "order": leading order of $V^\star$ as $\delta\to0$; "slope": $\log_2(V(0.2)/V(0.1))$ of the search; "n-dep.": the largest $|V_n/V_2-1|$ for $n=3,4$ and $\delta\ge0.1$ (for item 7 it reflects search shortfalls, for item 13 a genuine increase with n; items 19 and 20 differ for odd n, Theorem 6.1). "n.r.": no violation resolved by the search (*full* run on $CPU(4threads)$).

(16). Item 15 decreases with n, because the identity part of $e^{P/2}e^Qe^{P/2}$ enters N; item 13 increases with n at large δ (on M_3 at $\delta=0.9$: 0.3048 against 0.2197 on M_2, both certified); items 12, 14 and 16 give the same values for $n=2,3,4$. In item 16 the best points have large $\|A\|$, $\|B\|$, and the supremum is approached as the scale grows.

(iii) *Dimension.* For items 9 and 10 the search gives the same values for $n=3,4$ as for $n=2$, to 4×10^{-3}. Item 6 depends on n: for $n=2$ the search never exceeds $\kappa-1$, but for $n\ge3$ it certifies larger violations, $V\ge0.847$ at $n=4$, $\delta=0.8$, where $\kappa-1=0.667$; a stored point gives $V>2.1$ at $n=3$, $\delta=0.9$, where $\kappa-1=1.29$. We only know $\kappa-1\le V_6\le h-1$.

(iv) *Extremisers.* Except for items 6 ($n\ge3$), 11 ($A=W^{-1}$) and 25 (A of rank two with eigenvalue ratio $\approx\delta^2/8$), the extremisers are rank-one operators or matrix units in the span of eigenvectors for a and b, and they do not commute with W.

(v) *Limits of the search.* The suprema of 6 and 7 are approached only as an operator becomes singular; gradient ascent reaches 0.924 and 0.796 of the bound at $\delta=0.5$, and finds no violation of item 6 for $\delta\le0.2$. Item 25 is resolved for $\delta\ge0.3$; the search reaches m_{25} to 10^{-3} for $\delta\ge0.7$ and does not exceed it.

(vi) *Direct sums.* On $M_2\oplus M_3$ with $W=1\oplus s1$ and $W=\mathrm{diag}(1,\ell)\oplus s1$, $s\in\{\frac14,4\}$, intra-block contrast $\delta\in\{0.3,0.5,0.7\}$: for central W the largest value found for the items that hold for every trace is 1×10^{-16}; for 11 and 24 the search reproduces the exact values for the fixed density

to 2×10^{-11}; with intra-block contrast no value exceeds the M_2 envelope (largest relative excess 1×10^{-16}), and where Theorem 8.1 gives equality the search reaches it to 10^{-3} in 63 of 72 cases.

11 Conclusions and open problems

For the 25 characterisations of the trace studied here, the violation of the inequality by a non-tracial functional φ_W is controlled by two Michelson contrasts, that of W and that of the operators: it vanishes if either is zero, and for Gardner's inequality and its relatives it is an explicit function of their product. The largest violation is found, with few exceptions, at projections or at two-level operators $\ell+(1-\ell)P$, as the vertex principle predicts where it applies. Testing a characterisation on projections and their affine images is therefore the efficient way to detect a non-tracial functional, and the size of the violation measures the contrast of its density. The exceptions are structural: thresholds in the contrast of the operators, a peak at the contrast of the density for the determinant inequality, shift invariance for the Peierls–Bogoliubov inequality, and, in dimension three, the spectral geometric mean and the generalised Powers–Størmer inequality. The contrast of composite operators follows simple rules (Section 9): rapidities add for products and means, angles add for Jordan products, and the commutator test of traciality fails by the product of the three contrasts of W, A and B.

Conjecture 11.1. $V_n=\kappa^2-1$ for item 7, $V_n=m(\delta)$ for item 10 and $V_n=m_{25}(\delta)$ for item 25 in every dimension; $V_2=\kappa-1$ for item 6; and the value of Proposition 5.4 holds for item 9 in every dimension.

Further questions: (0) proofs of the threshold phenomenon of Section 10.4(b), i.e. of inequalities that hold for every φ_W as long as the operators have contrast below a critical value depending on $\Delta(W)$, and of the equality in Proposition 5.3; (1) the envelope of item 6 for $n\geq3$, where the search exceeds $\kappa-1$, and whether it grows with n; (2) closed forms for items 12–16, whose leading coefficients the search gives as $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{36}$, 0.064 and 0.09 (item 14: the leading coefficient $(1-\frac{2}{p})^2/4$ is a lower bound); (3) the analogues for infinite-dimensional von Neumann algebras, where the contrast of an unbounded density may be 1 while the functional is still close to a trace in other senses; (4) envelopes for other inequalities that characterise traces, e.g. those of [27, 15] with the arithmetic mean, and for the dual problem of characterising central elements [28].

Code availability. The library, the programs that compute all numbers, tables and figures, and the tests are available at `https://github.com/andre-a-no/von_neumann_type_i` (MIT licence).

A Random samples and certified envelopes, item by item

Each figure belongs to one inequality and has one panel per dimension. Top row: the normalised violation $(L-R)/N$ of the 200000 random samples (grey) and, where the variables are not projections, of the 200000 scaled-projection samples (orange) against $\Delta(W)$ on M_2, M_3, M_4 (Section 10.2); black: the certified envelope $V(\delta)$ on the grid of Section 10, joined by straight lines. Bottom row (items with operator variables): random samples with $\Delta(W) = 0.9$ and every operator variable of one contrast δ', against δ' on M_2 and M_3; black: the certified $V(0.9, \delta')$ of Section 10.4. Samples far below zero are cut off. Where a curve is concave between grid points (item 11 below its maximum at $\delta' = \delta$, and after its steep drop near $\delta' = 1$) a few samples lie above the straight segments; on M_2 they lie below the closed form $1 - (w_1\ell' + w_2)/(2\sqrt{\ell'})$, $\ell' = \frac{1-\delta'}{1+\delta'}$, $w = \frac{2}{1+\ell}(1, \ell)$.

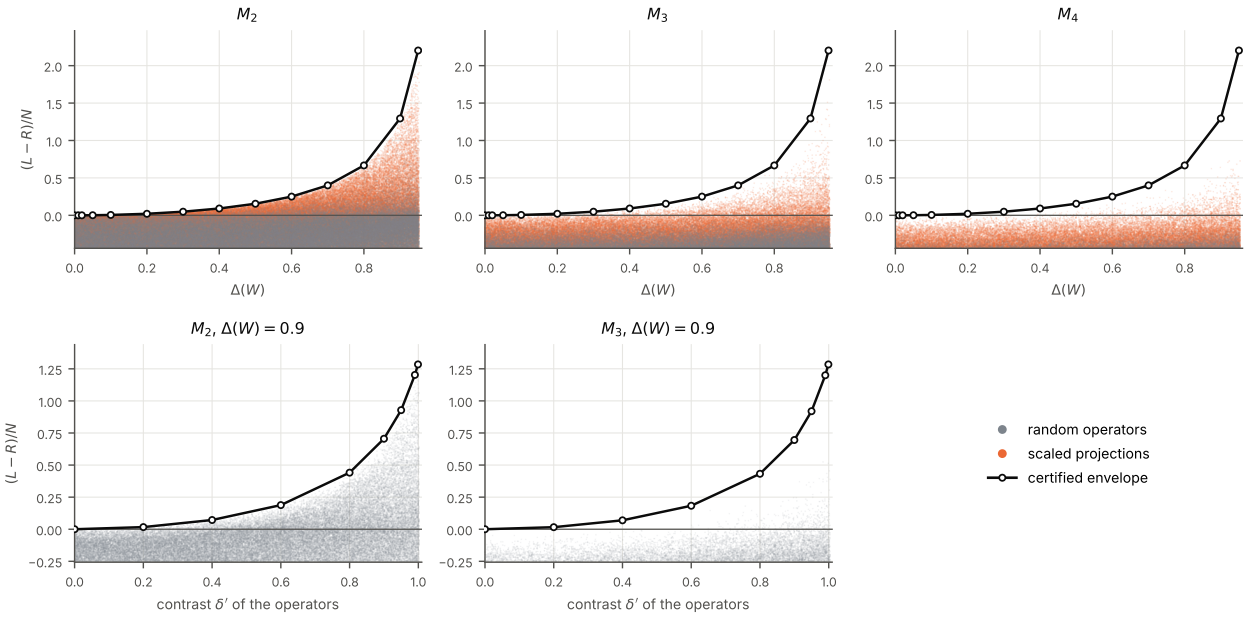


Figure 9: Item 1, $|\varphi(X)| \leq \varphi(|X|)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

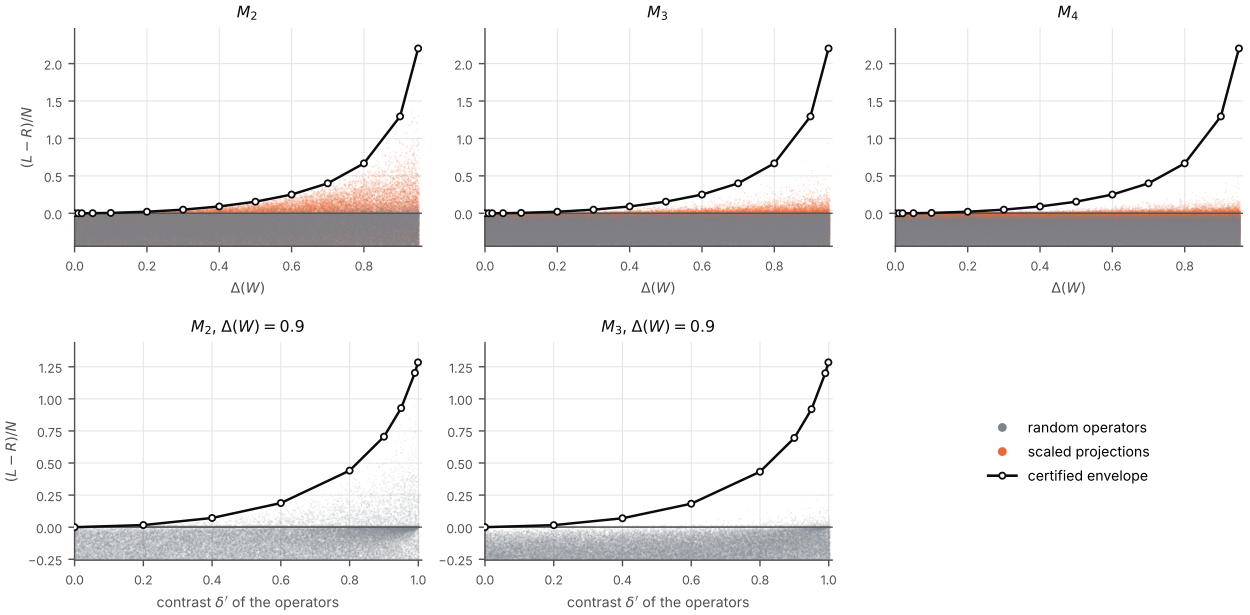


Figure 10: Item 2, $\varphi(|X+Y|) \leq \varphi(|X|) + \varphi(|Y|)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

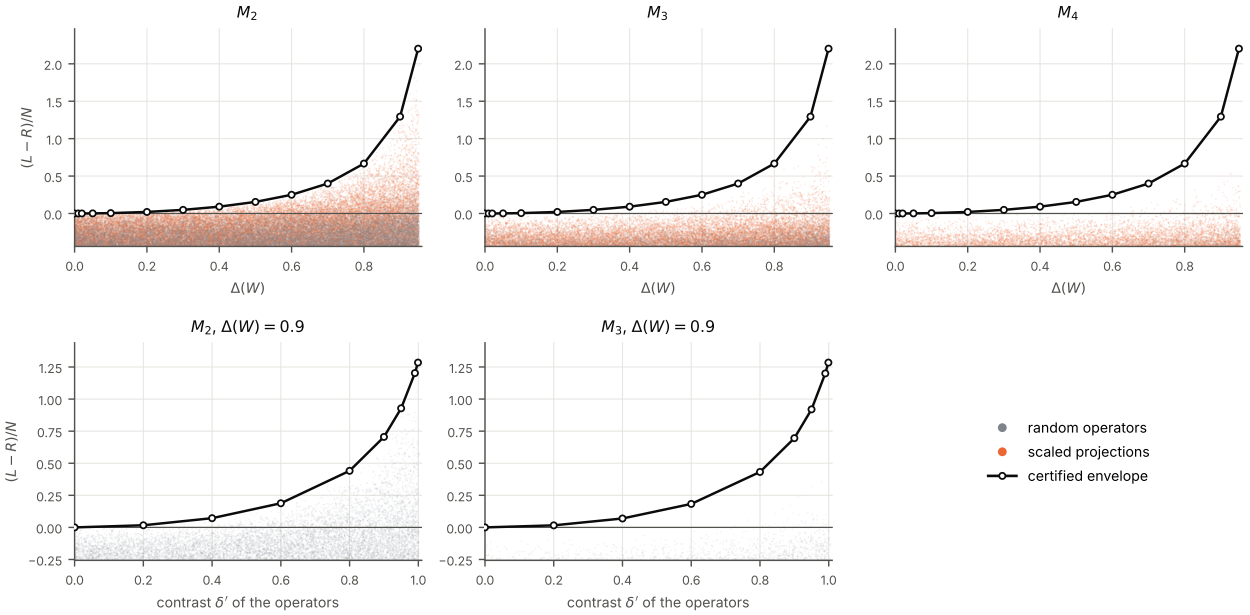


Figure 11: Item 3, $\varphi(A + A^*) \leq 2\varphi(|A^*|)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

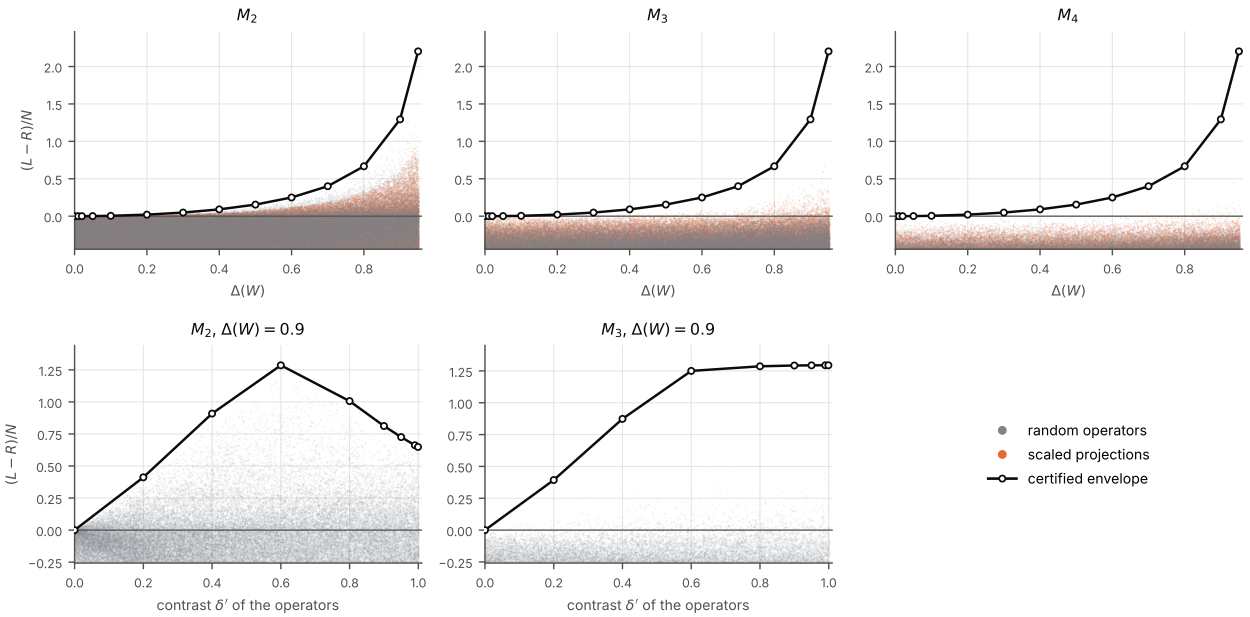


Figure 12: Item 4, $|\varphi(A^2)| \leq \varphi(A^*A)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

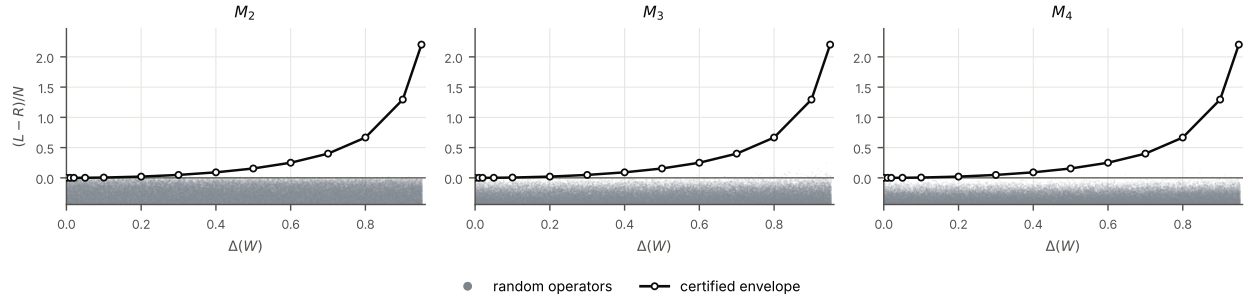


Figure 13: Item 5, $\varphi(|P-R|) \leq \varphi(|P-Q|) + \varphi(|Q-R|)$: random samples and the certified envelope against $\Delta(W)$.

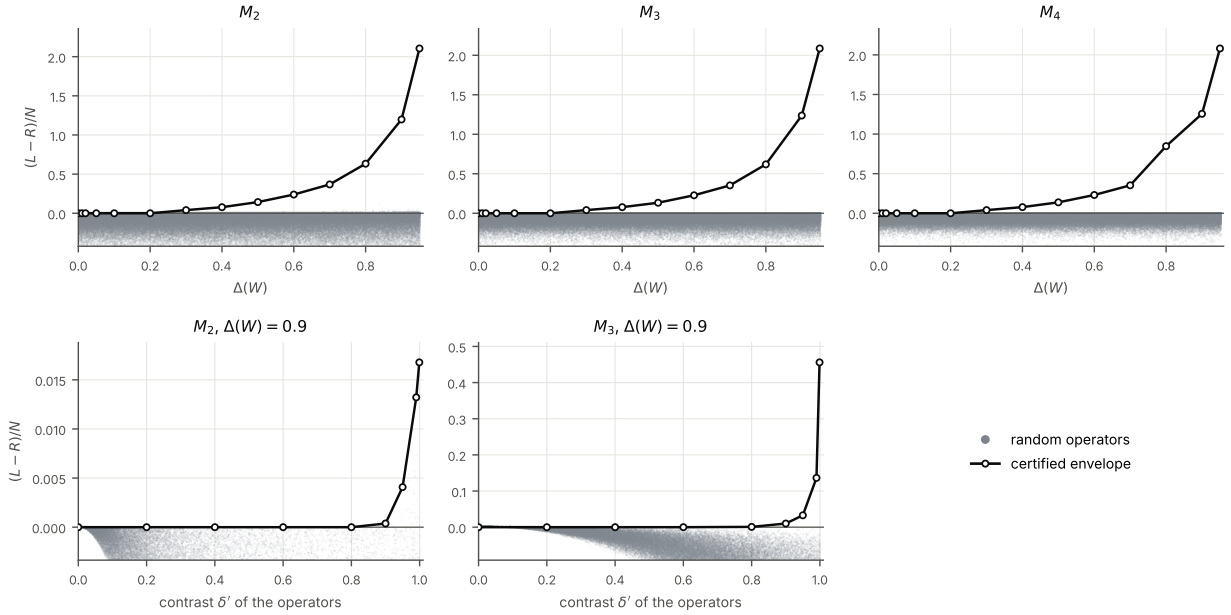


Figure 14: Item 6, $\varphi(A \natural B) \leq \sqrt{\varphi(A)\varphi(B)}$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

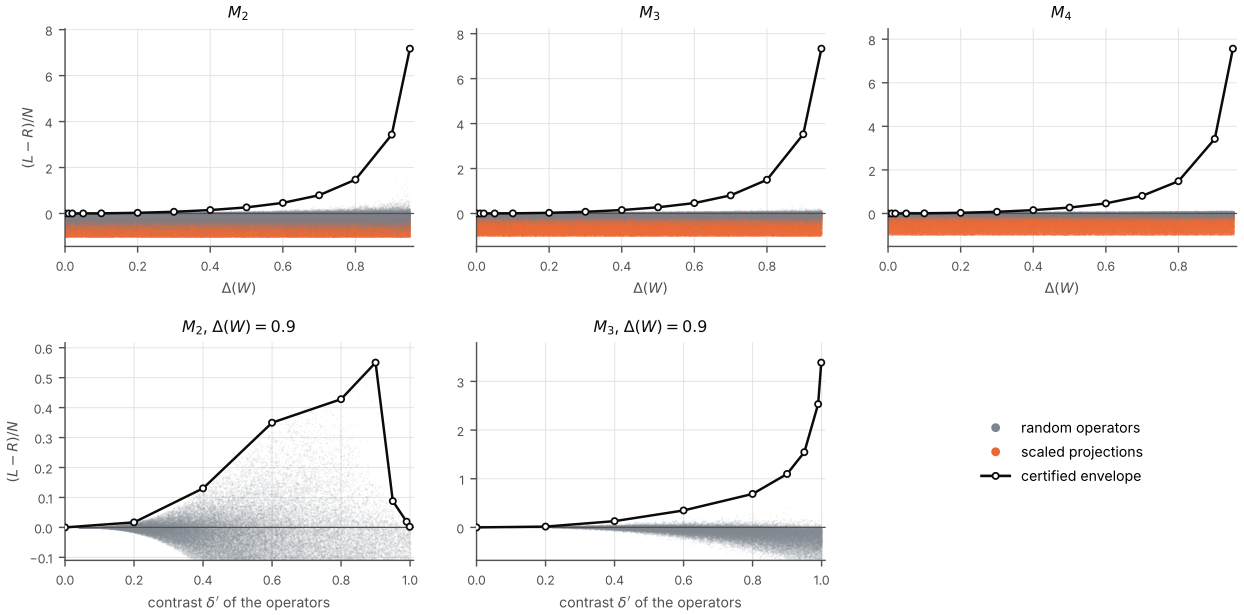


Figure 15: Item 7, $\varphi((B^{1/2}AB^{1/2})^2) \leq \varphi(BA^2B)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

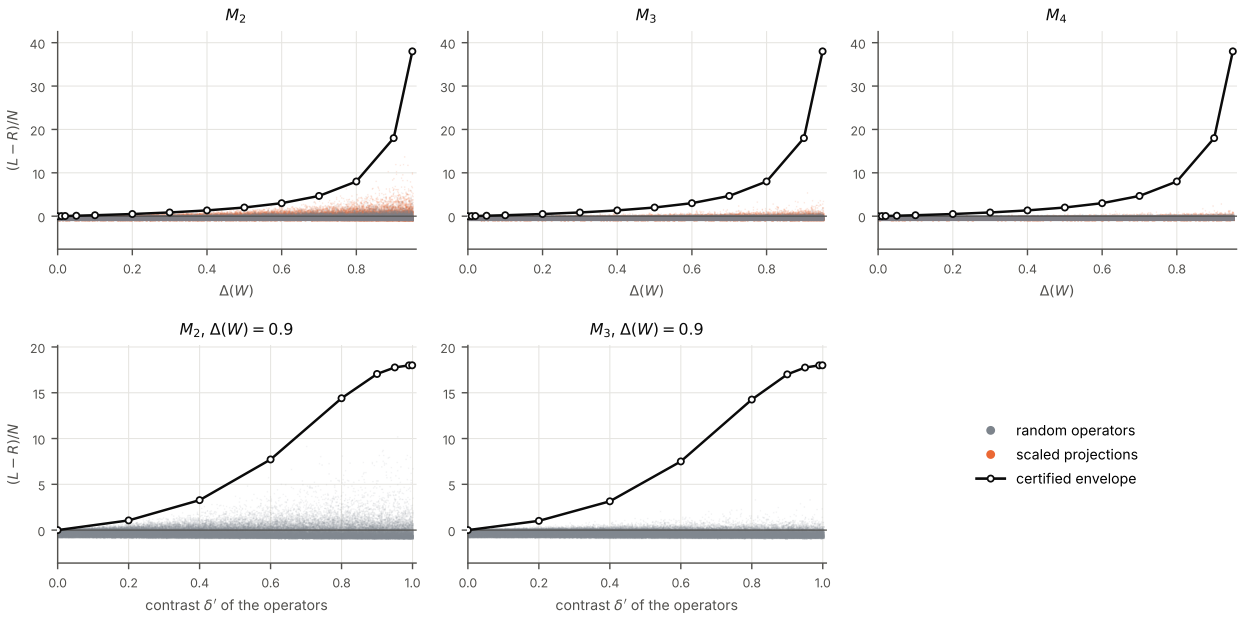


Figure 16: Item 8, $\varphi(|AB^*|) \leq \frac{1}{2}\varphi(|A|^2) + \frac{1}{2}\varphi(|B|^2)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

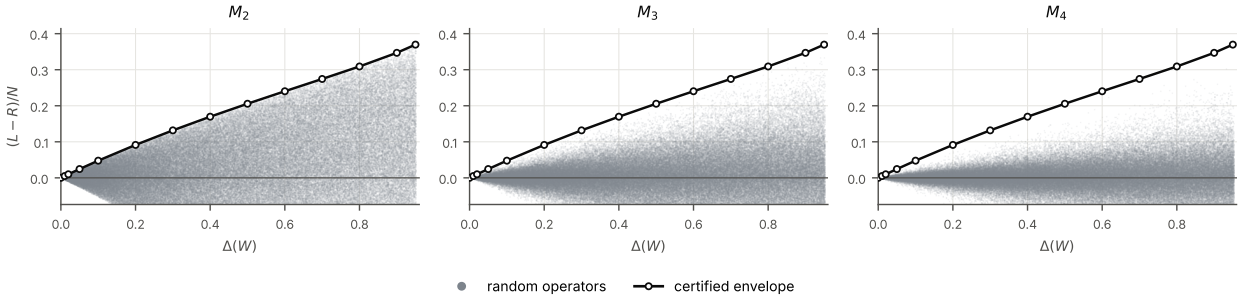


Figure 17: Item 9, $\varphi(PQ + QP) \leq 2\varphi(QPQ)$: random samples and the certified envelope against $\Delta(W)$.

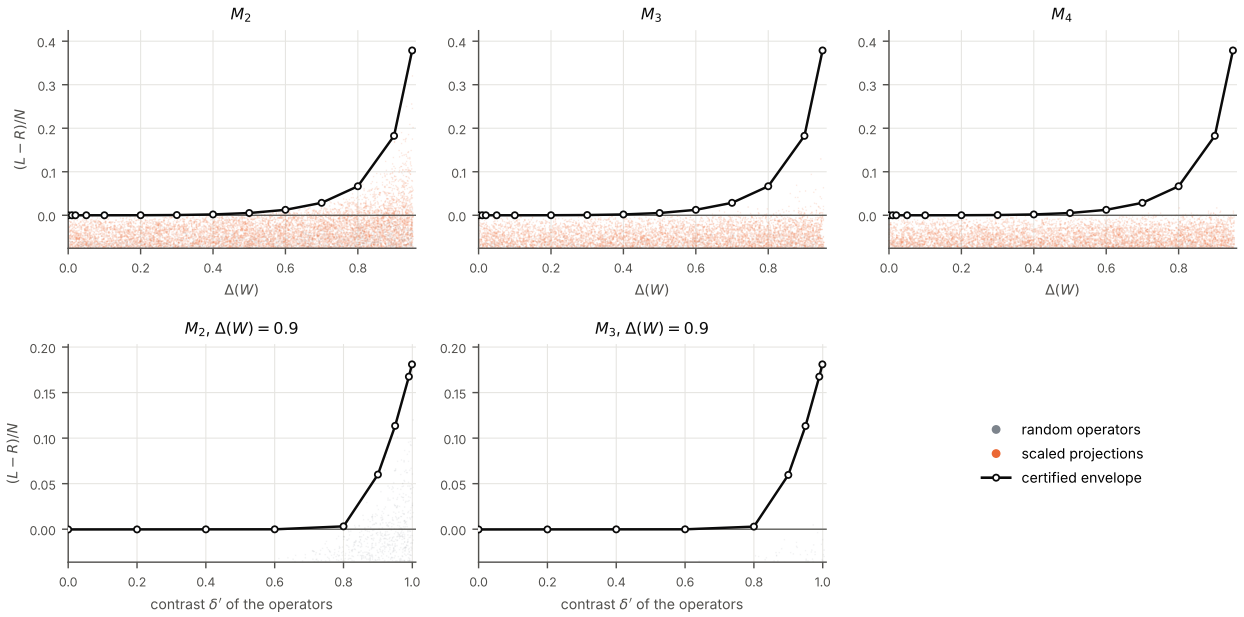


Figure 18: Item 10, $0 \le A \le B \Rightarrow \varphi(A^2) \le \varphi(B^2)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

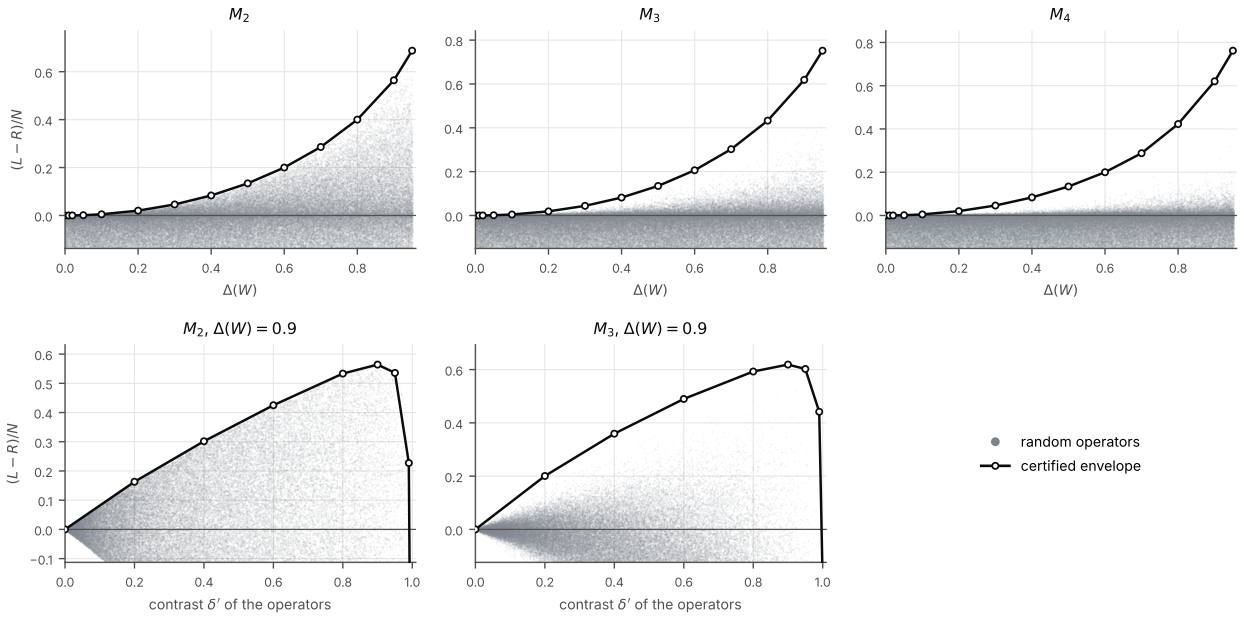


Figure 19: Item 11, $(\det A)^{1/n} \le \frac{1}{n}\varphi(A)$, $\varphi(1) = n$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

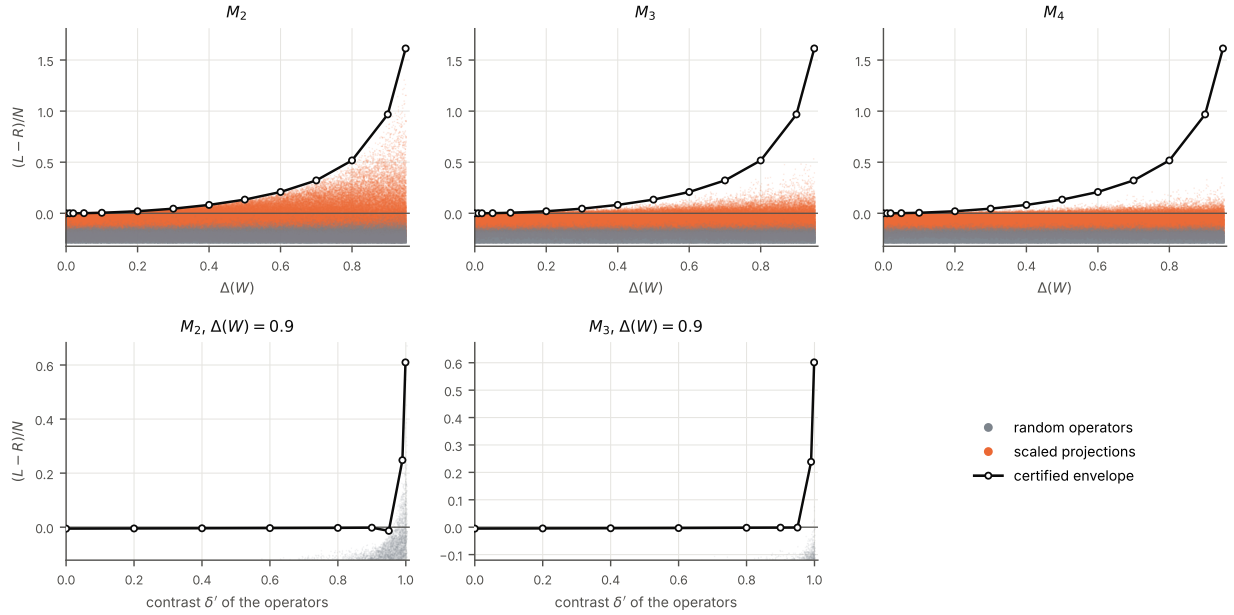


Figure 20: Item 12, $\varphi((A+B)^{1/2}) \leq \varphi(A^{1/2}) + \varphi(B^{1/2})$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

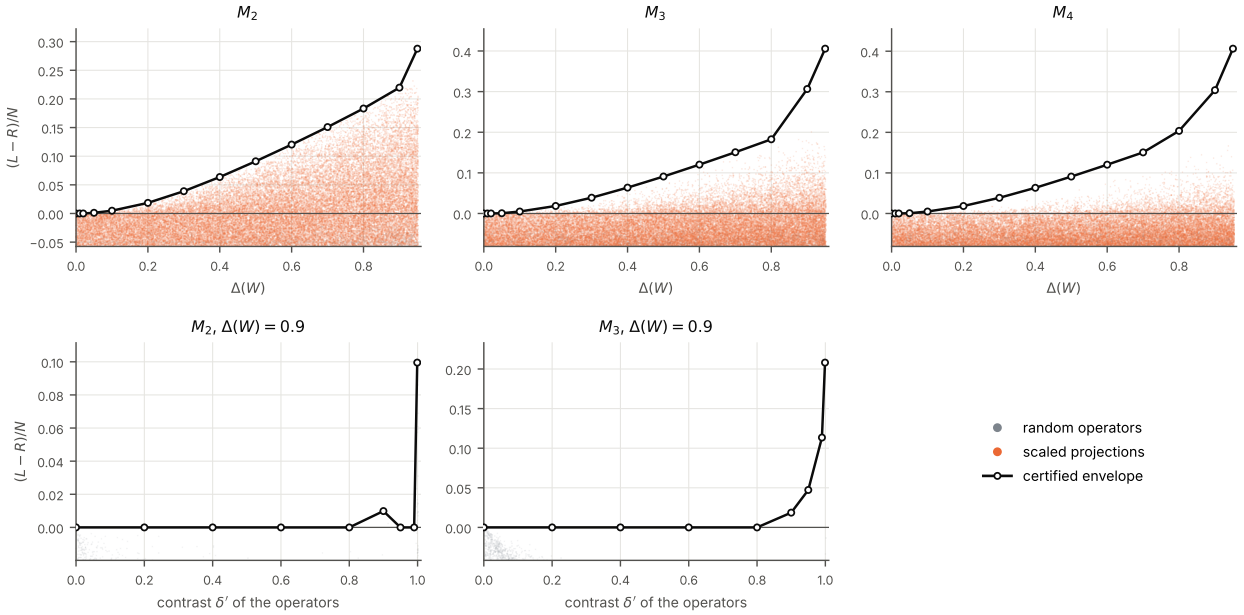


Figure 21: Item 13, $\varphi(A + B - |A - B|) \leq 2\varphi(A^{1/4}B^{1/2}A^{1/4})$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

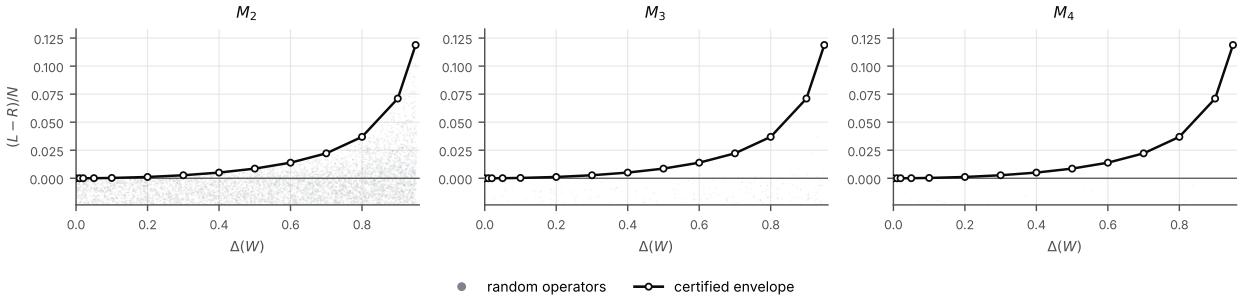


Figure 22: Item 14, $|\varphi(PQ)| \leq \varphi(P)^{1/3}\varphi(Q)^{2/3}$: random samples and the certified envelope against $\Delta(W)$.

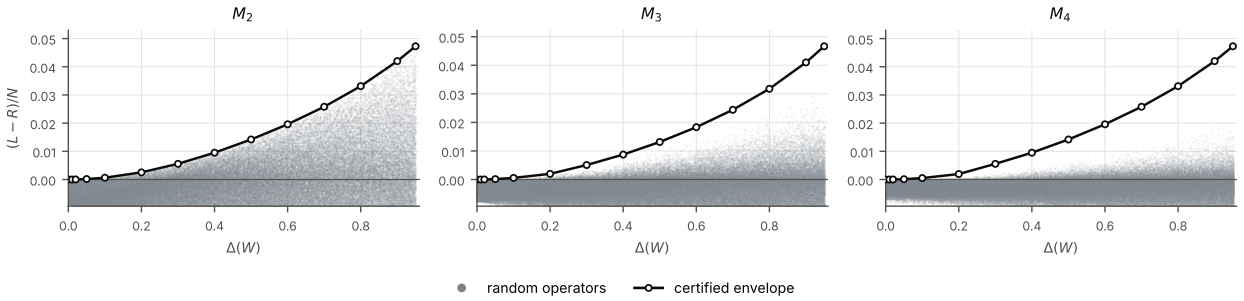


Figure 23: Item 15, $\varphi(e^{P+Q}) \leq \varphi(e^{P/2}e^{Q}e^{P/2})$: random samples and the certified envelope against $\Delta(W)$.

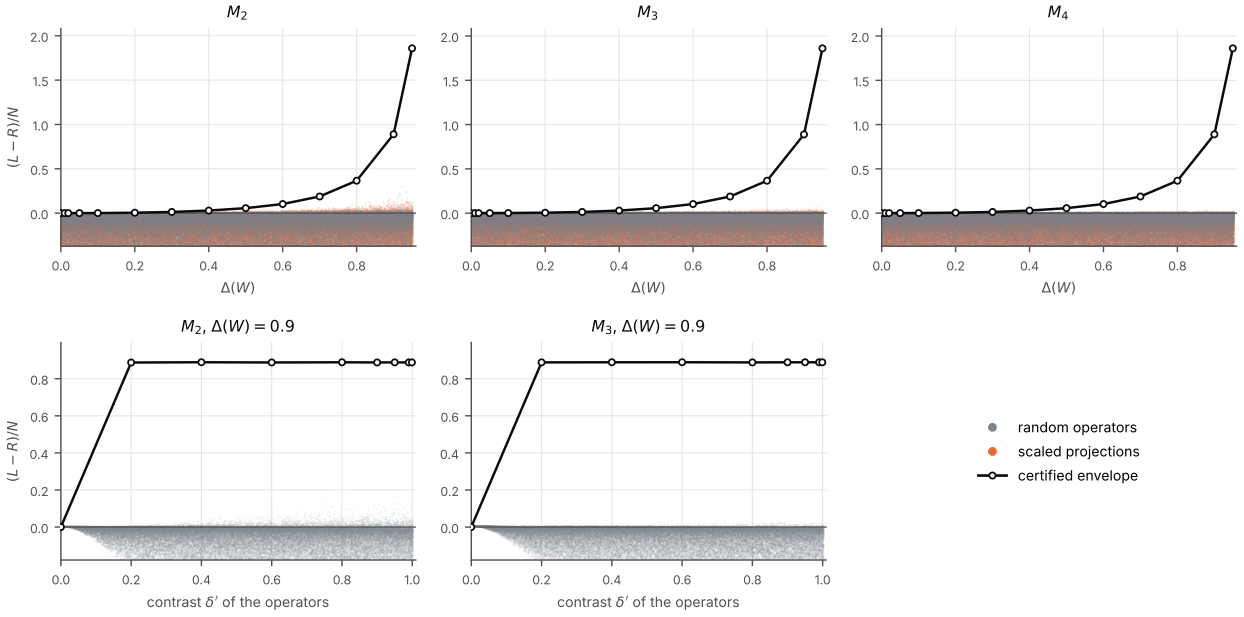


Figure 24: Item 16, $\varphi(e^{A})e^{\varphi(e^{A/2}Be^{A/2})/\varphi(e^{A})} \leq \varphi(e^{A+B})$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

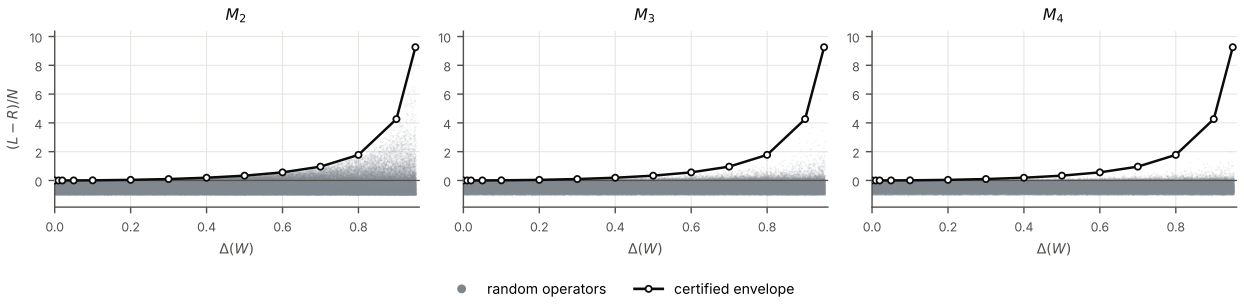


Figure 25: Item 17, $\varphi(PQP) \leq \varphi(Q)$: random samples and the certified envelope against $\Delta(W)$.

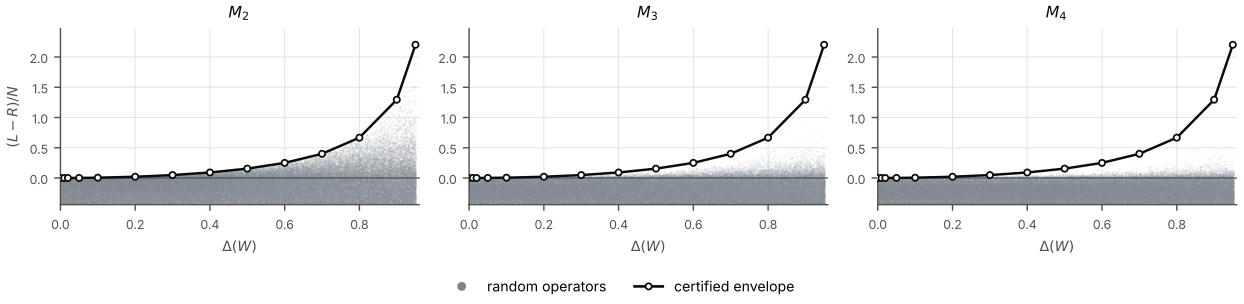


Figure 26: Item 18, $\varphi(|P-Q|) \leq \varphi(P+Q)$: random samples and the certified envelope against $\Delta(W)$.

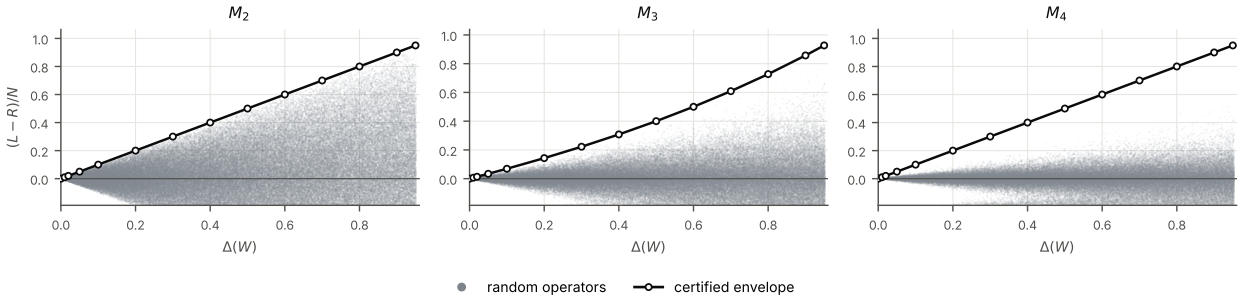


Figure 27: Item 19, $\varphi(P+Q) \leq \varphi(1+P \wedge Q)$: random samples and the certified envelope against $\Delta(W)$.

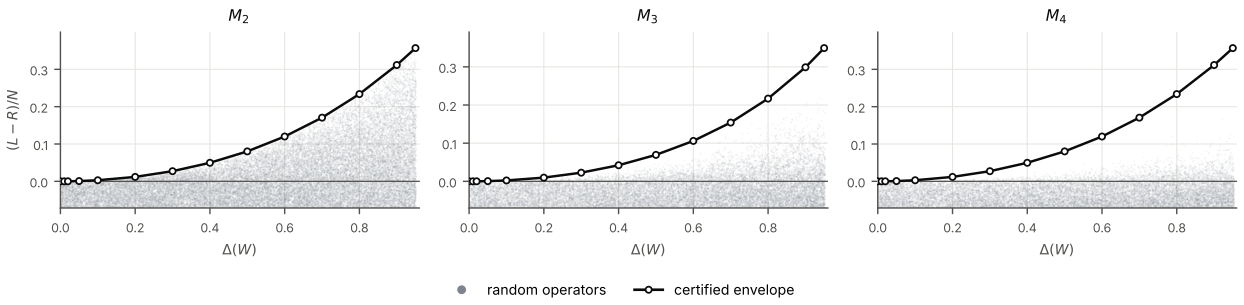


Figure 28: Item 20, $\varphi(e^{QPQ}) \leq \varphi(e^{P})$: random samples and the certified envelope against $\Delta(W)$.

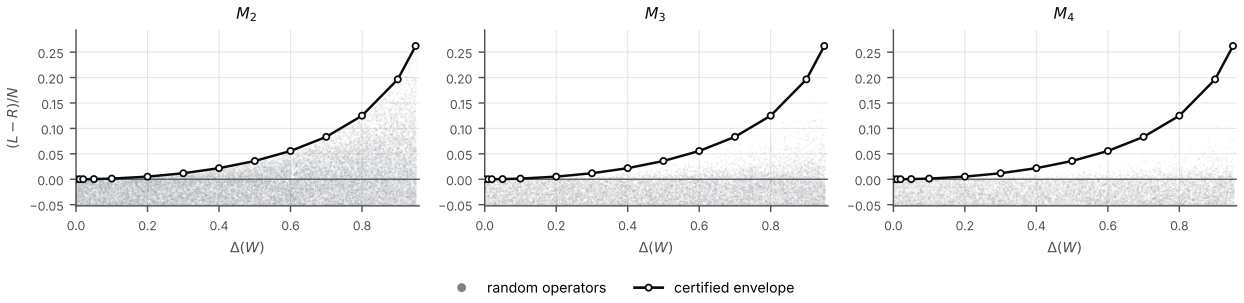


Figure 29: Item 21, $\varphi(\mathrm{Re}\, PQ) \geq 0$: random samples and the certified envelope against $\Delta(W)$.

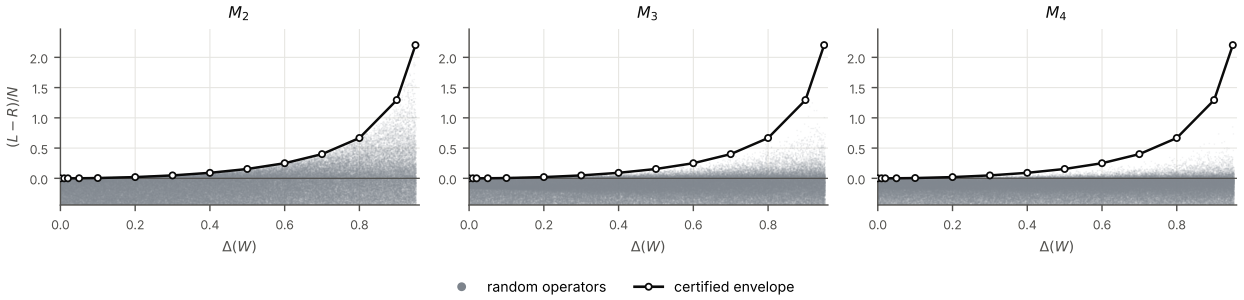


Figure 30: Item 22, $\varphi(\mathrm{Re}\, PQ) \leq \varphi(|PQ|)$: random samples and the certified envelope against $\Delta(W)$.

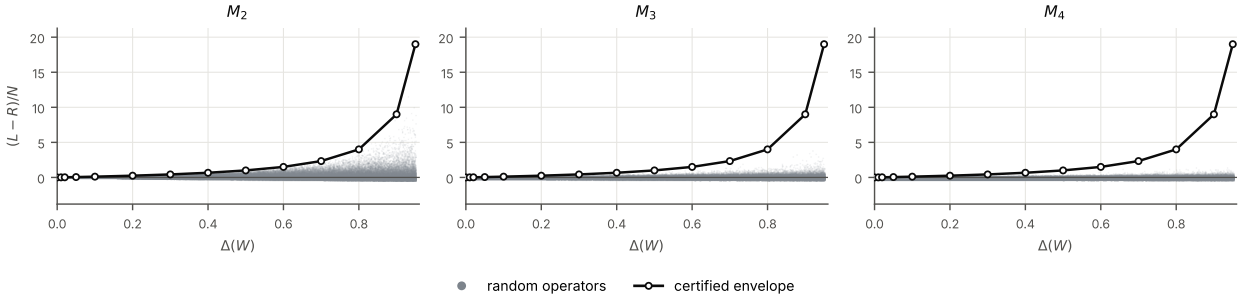


Figure 31: Item 23, $\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$: random samples and the certified envelope against $\Delta(W)$.

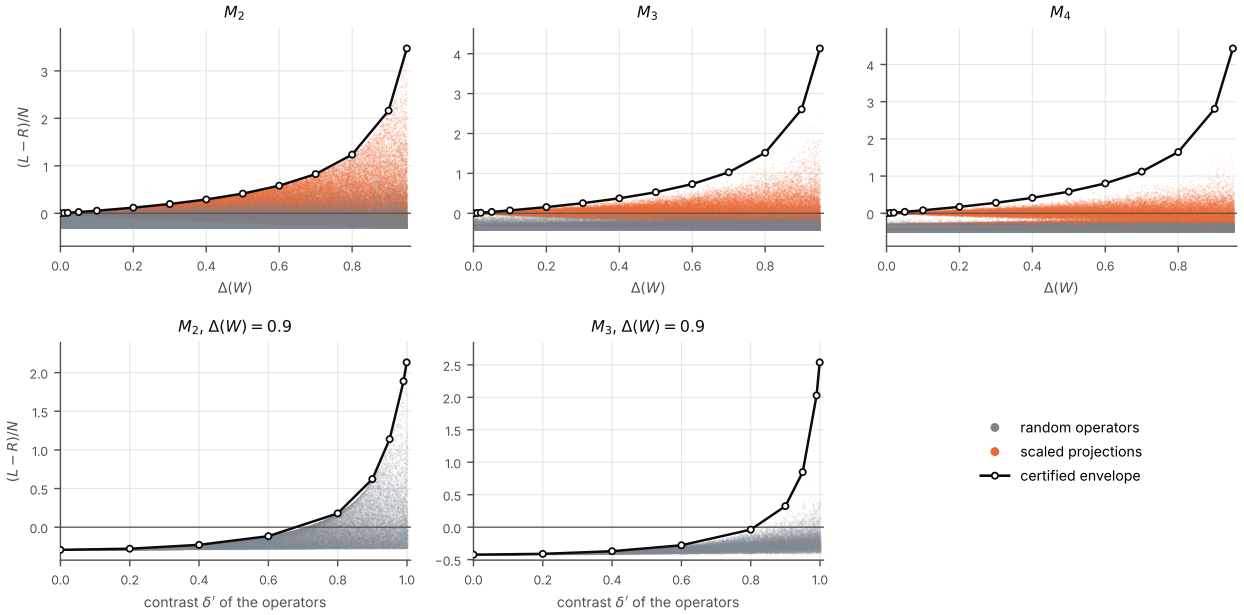


Figure 32: Item 24, $\varphi(A^2)^{1/2} \leq \varphi(A)$, $\varphi(1) = n$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

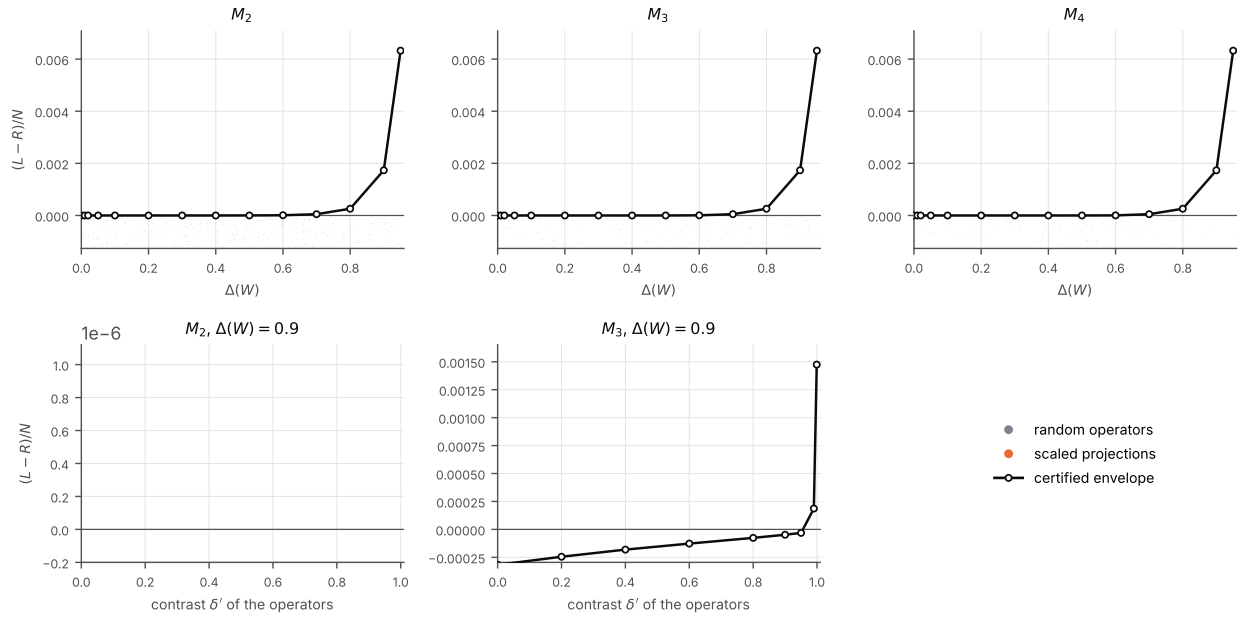


Figure 33: Item 25, $0 \leq A \leq B \Rightarrow \varphi(A^4) \leq \varphi(AB^2A)$: random samples and the certified envelope; top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

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